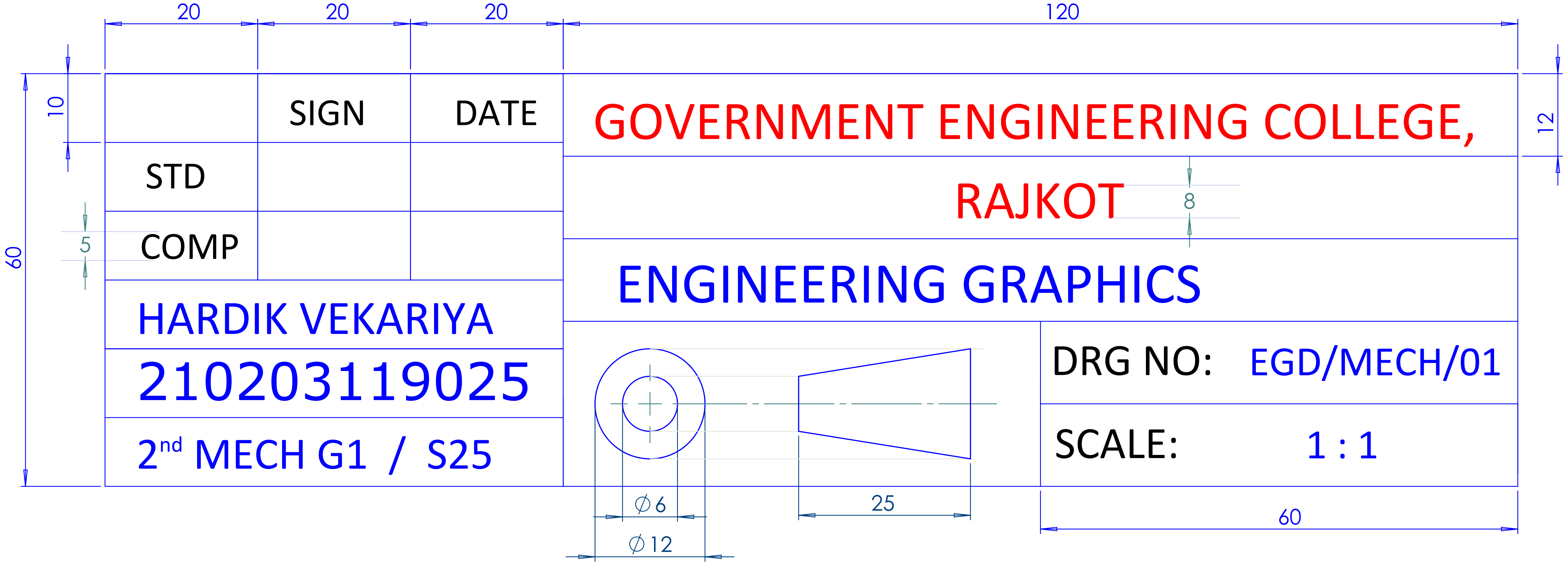
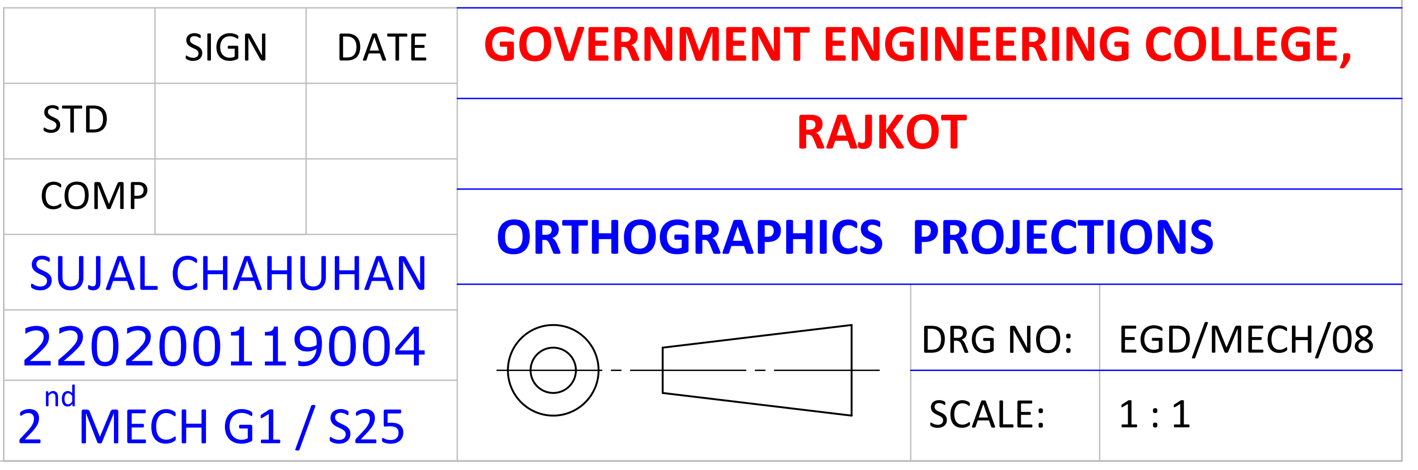
# TITLE BLOCK




TITLE BLOCK should be drwan to the scale (dimension as shown above) in each sheet,
it should touch boundry of lower left corner.
Do not show dimensions.

ALL DIMENSIONS ARE IN MM



# 1: Practice Sheet

Q.1. Shows Different types of lines and their applications.

Q.2. With suitable drawing show Dimensioning method.

Q.3. Dividing the circle of 60mm diameter in to 8 and 12 equal parts.

Q.4. Dividing long line in to number of equal parts.
[Take 92 mm long line and divide it in 7 equal parts]

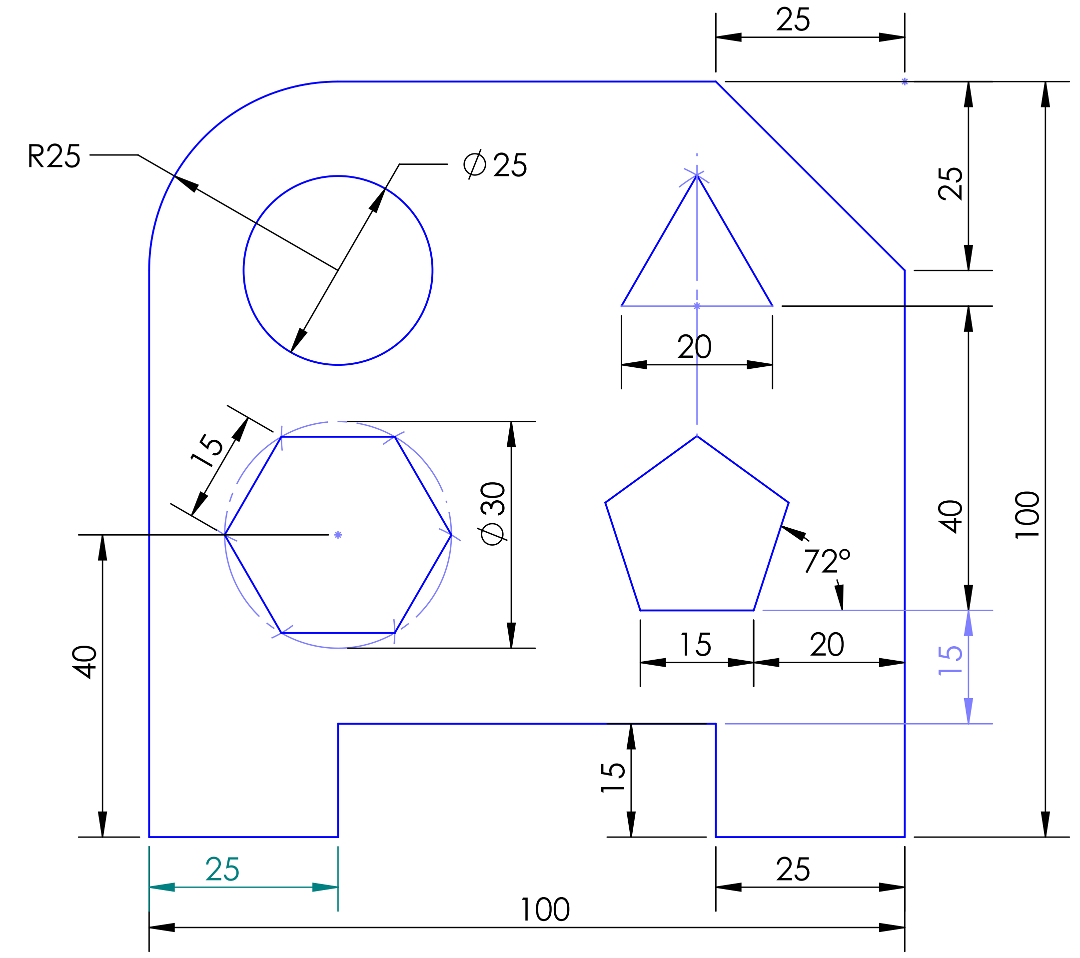
Q.5. Bisecting Line [Take 65mm long line and angle 30°]

Q.6. Draw Polygon by Universal Method. [Side of polygon = 40mm]

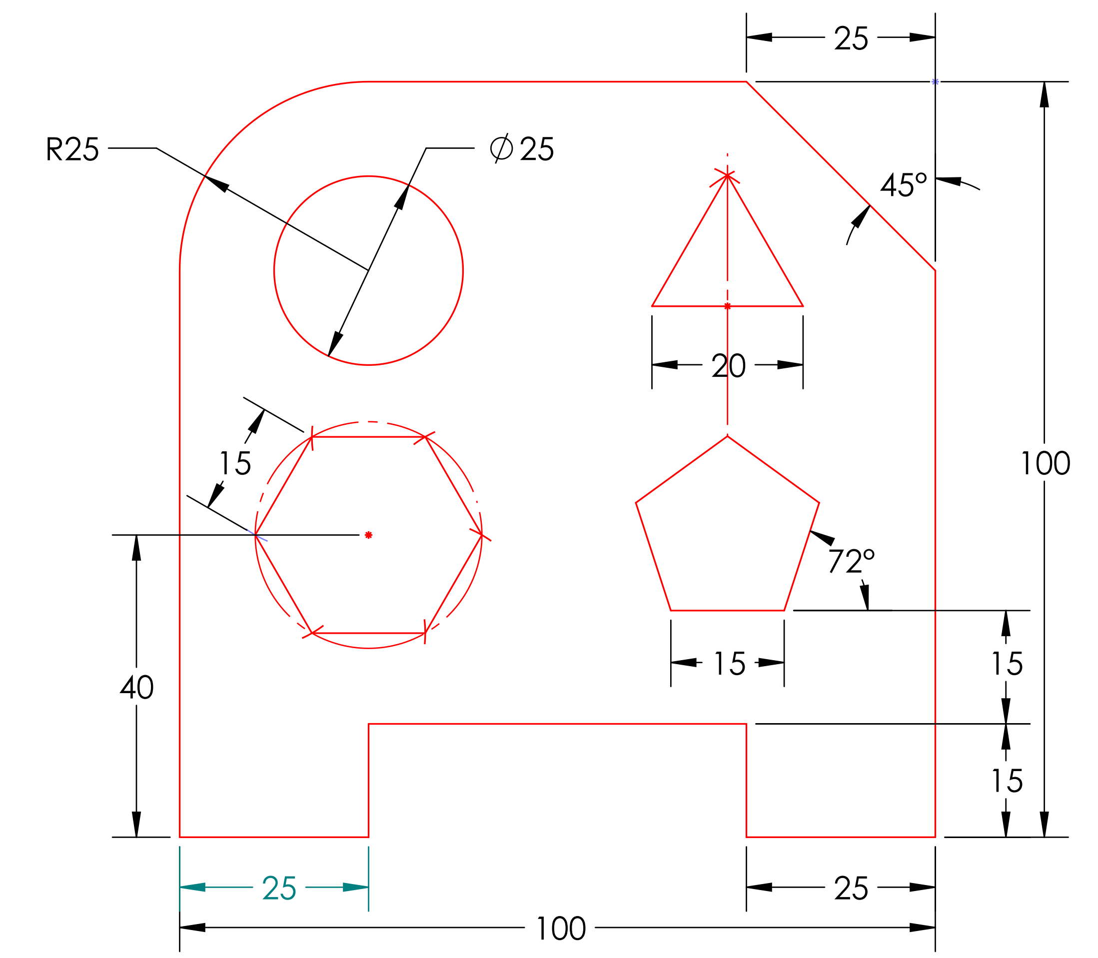
## 1 LINE TYPES

| LINES | ILLUTRATION | APPLICATIONS |
|---|---|---|
| CONTINIOUS THICK 0.5mm | | VISIBLE OUT LINE |
| CONTINIOUS THIN 0.2mm | | DIMENSION, PROJECTIONS CONSTRUCTIO, HATCH |
| DASHESD LINE | | HIDDEN OBJECT |
| CHAIN THIN | | CENTER LINE, SYMMETRY PICH CIRCLE LINE |
| CHAIN THIN WITH THICK ENDS | | CUTTING PLANE LINE |
| THIN WAVY | | IRREGULAR BOUNDRY |
| SHORT ZIGZAG | | BREAK LINES |
| ARROW | | DIMENSION, ANNOTATION |

## 2 METHOD OF DIMENSIONING

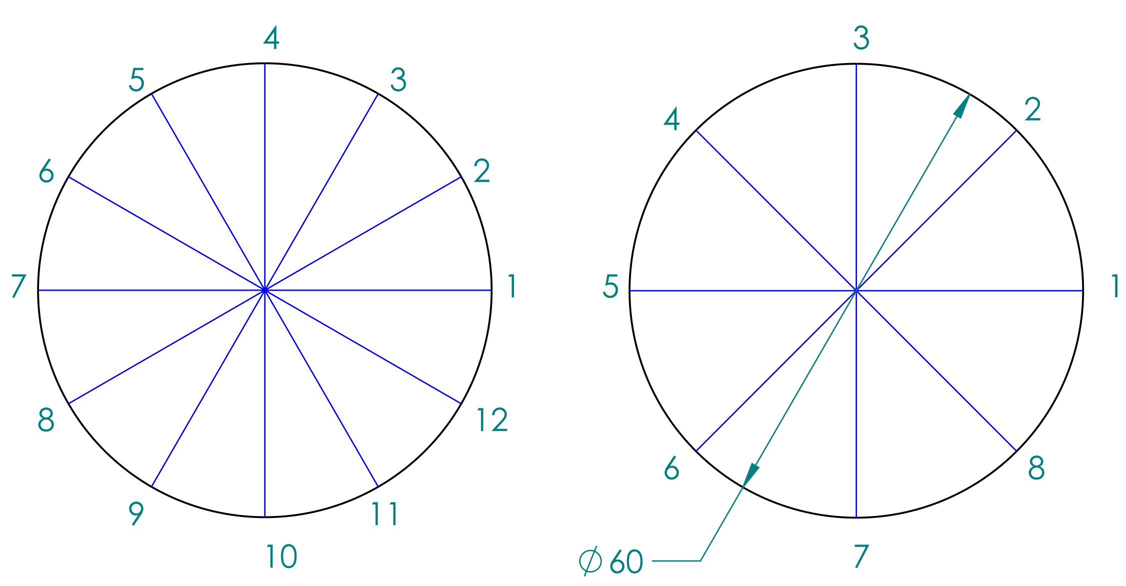



ALLIGN

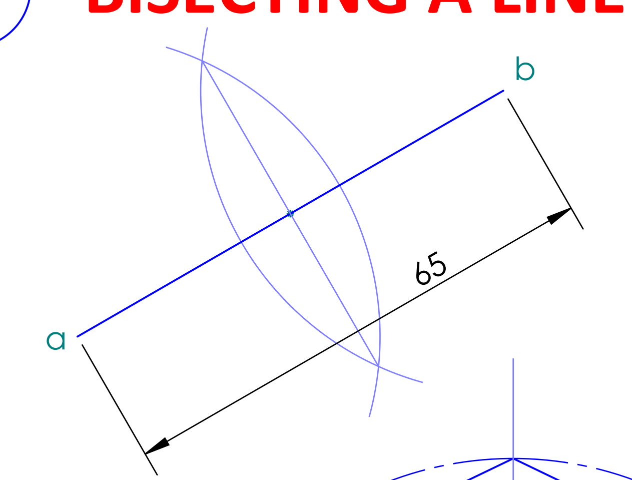



UNIDIRECTIONAL

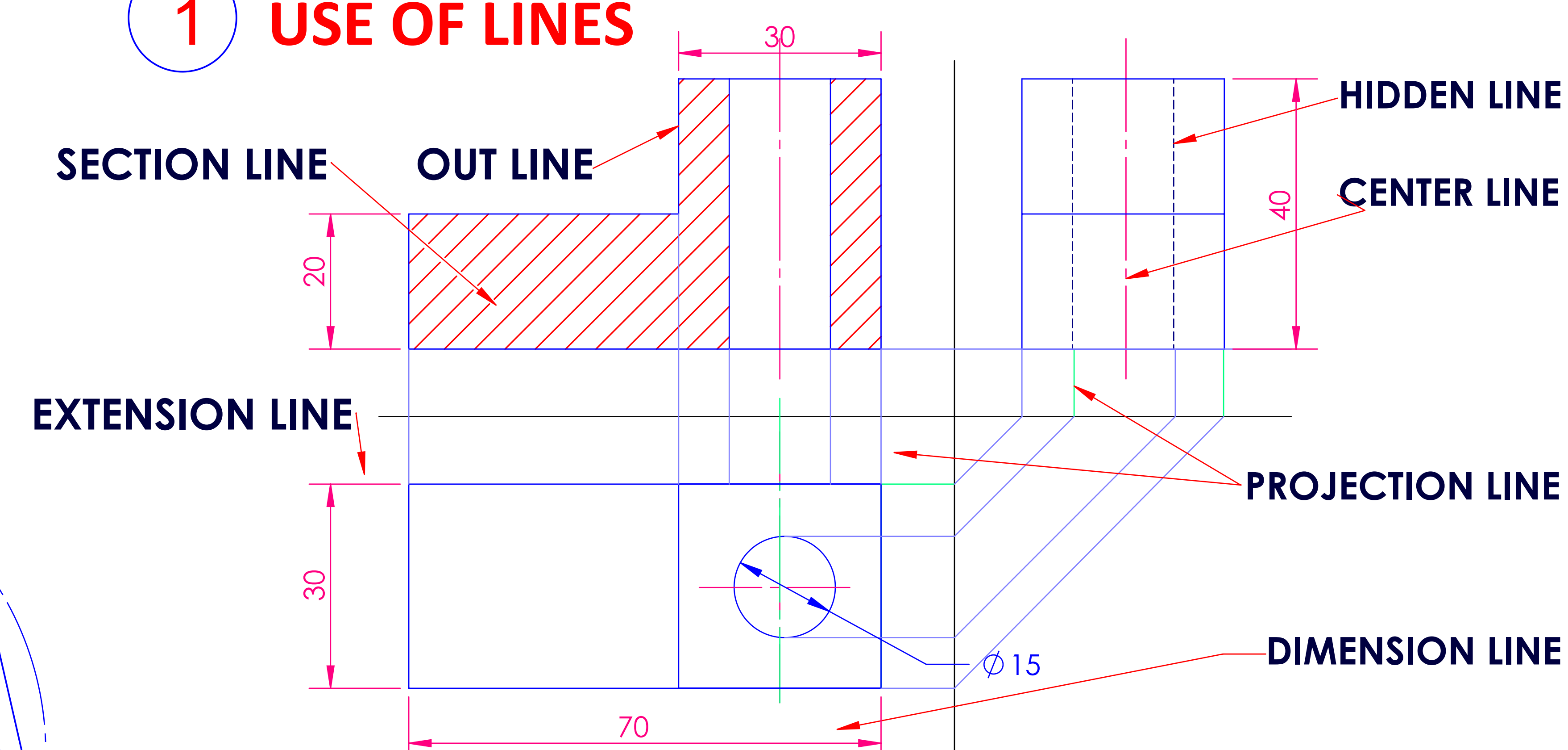
## 3 DIVISION OF CIRCLE

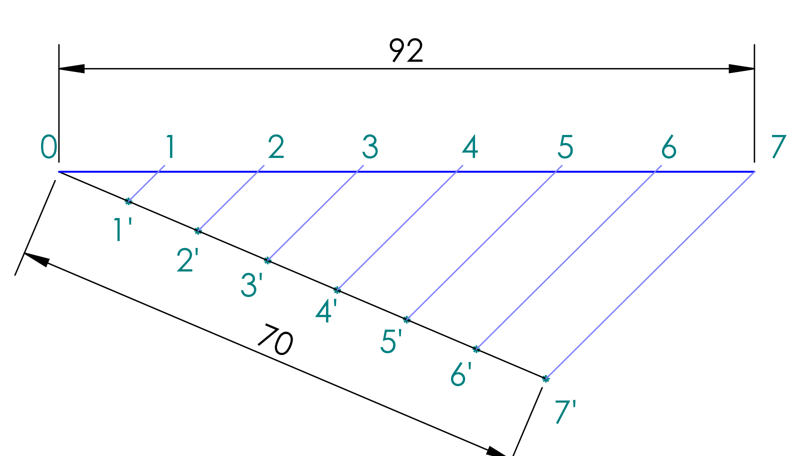



## 5 BISECTING A LINE

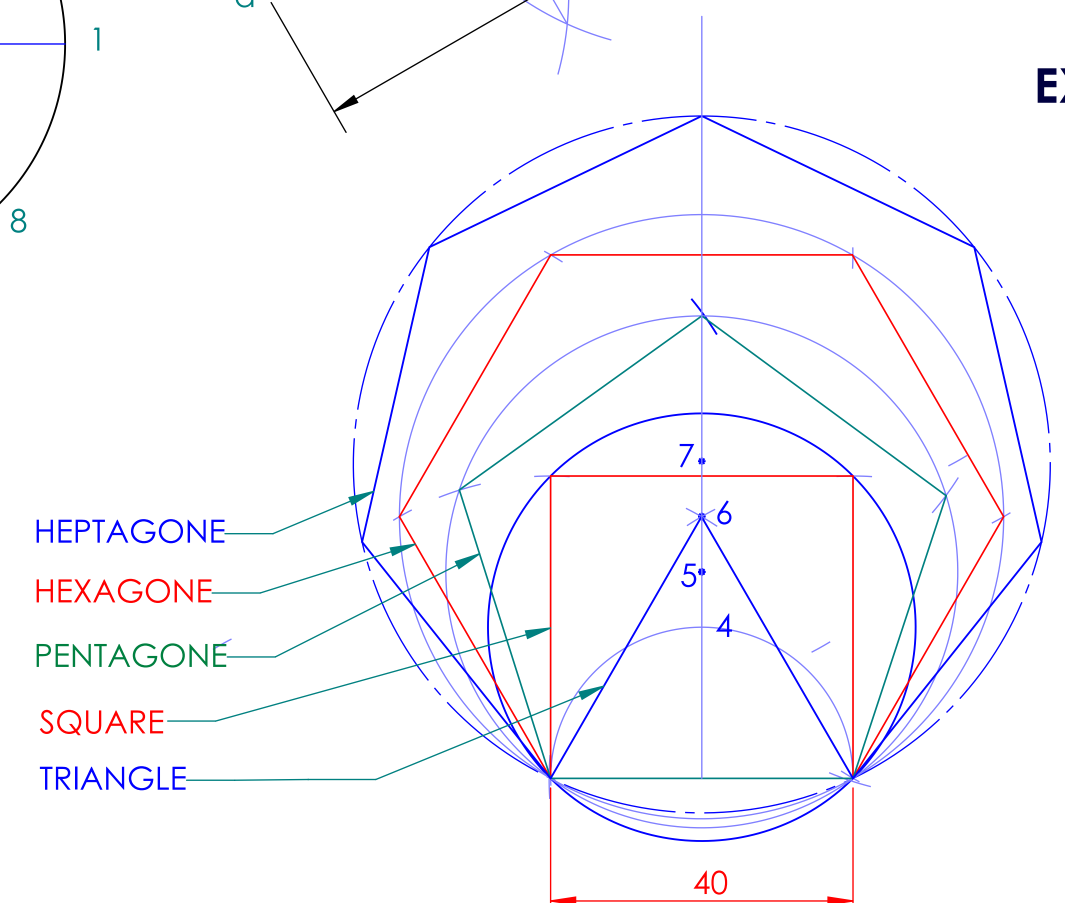



## 1 USE OF LINES




## 4 DIVISION OF LINE




## 6 UNIVERSAL POLYGONE

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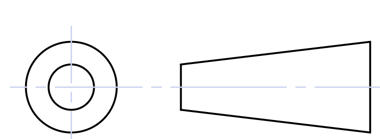
|---|---|---|---|---|
| STD | | | | |
| COMP | | | PRACTICE SHEET | |
| ROHAN PARMAR | | |  | DRG NO: EGD/MECH/01 |
| 200200119015 | | | | |
| 2nd MECH M4/ S21 | | | | SCALE: 1 : 1 |

## 2 : Plains scale and Diagonal Scale

*PROBLEM NO.1:*

*Construct a scale of 1:4, to show centimeters and long enough to measure up to 5 decimeters and show 3.7 decimeters on it.*

*PROBLEM NO.2:*

*A length of 4.5 meters is represented by a 3 cm long line. Extend this line to measure up to 30 meters. Show the length of 22 meters on this line.*

*PROBLEM NO.3:*

*On a map, the distance between two points is 14 cm. The real distance between them is 20 km. Draw a diagonal scale of this map to read kilometers and hectometers, & to measure up to 25 km. Show a distance of 17.6 km on this scale.*

*PROBLEM NO.4:*

*An area of 144 sq. cm on a map represents an area of 36 sq. km on the field. Find the of R.F. the scale and draw a diagonal scale to show kilometers, hectometers and decameters and to measure up to 10 kilometers. Indicate on a distance of 7 kilometers, 5 hectometers and 6 decameters on the scale.*

## PLAIN SCALE

**1**

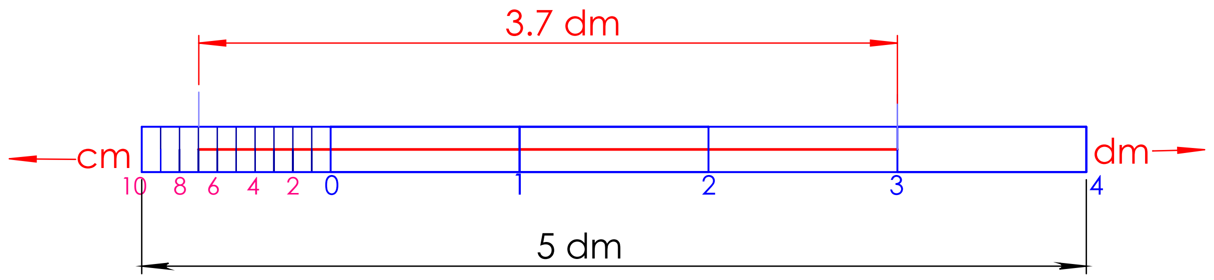
$$R.F. = \frac{\text{Length of scale}}{\text{Actual length}} = \frac{1}{4}$$

Length of scale = R F x Max length to measure

= 0.25 x 5 dm

= 0.25 x 5 x 100 mm

=125 mm



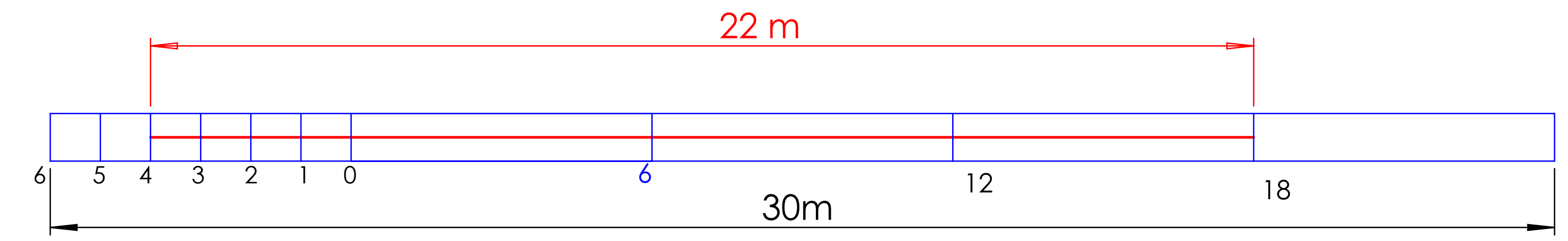

R.F. = 1 : 4

**2**

$$R.F. = \frac{\text{Length of scale}}{\text{Actual length}} = \frac{3\text{ cm}}{4.5\text{ m}} = \frac{3 \times 10\text{ mm}}{4.5 \times 1000\text{ mm}} = \frac{1}{150}$$

Length of scale = R F x Max length to measure

$$= \frac{1}{150} \times 30\text{ m} = \frac{1}{150} \times 30 \times 1000\text{ mm} = 200\text{ mm}$$




R.F. = 1 : 150

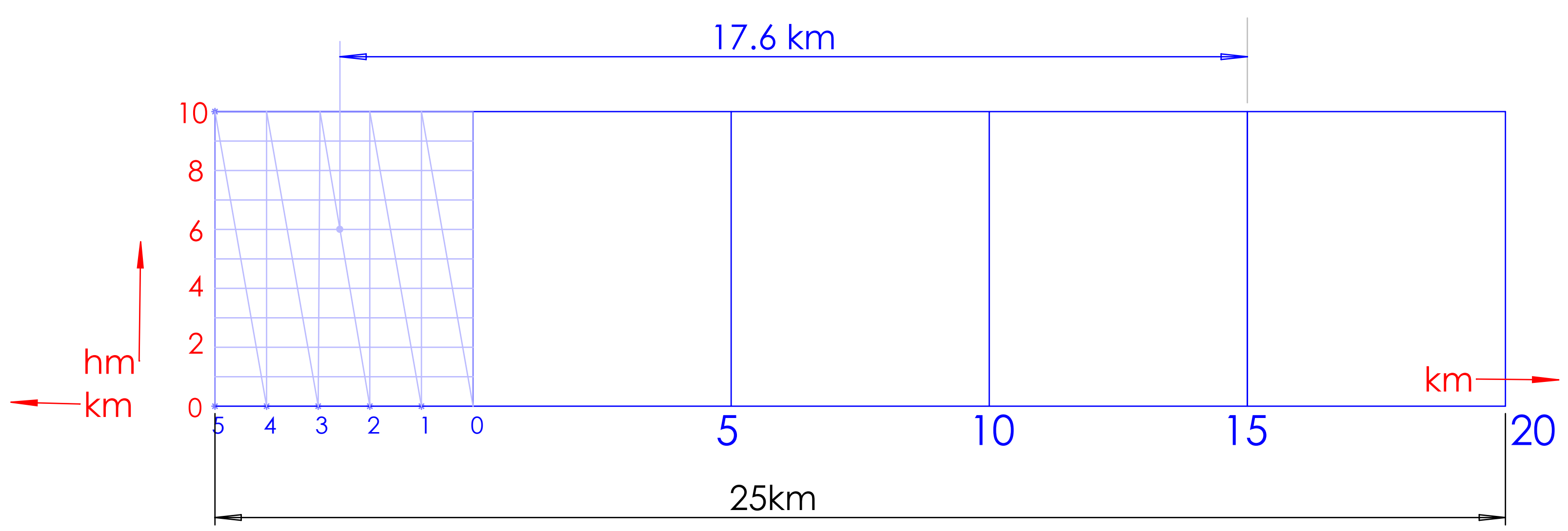
## DIAGONAL SCALE

**3**

$$R.F. = \frac{\text{Length of scale}}{\text{Actual length}} = \frac{14\text{ cm}}{20\text{ km}} = \frac{14 \times 10\text{ mm}}{20 \times 10^3 \times 10^3\text{ mm}} = \frac{7}{10^6}$$

Length of scale = R F x Max length to measure

$$= \frac{7}{10^6} \times 25 \times 10^6 = 175\text{ mm}$$



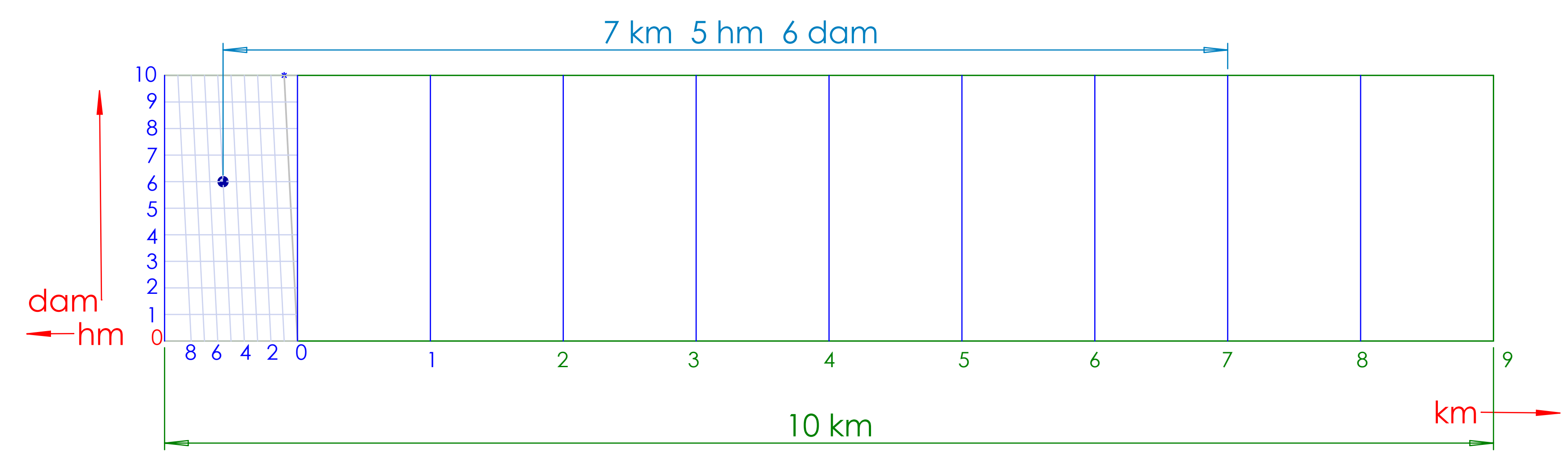

R.F. = 7 : $10^6$

**4**

$$R.F. = \frac{\text{Area of scale}}{\text{Actual area}} = \sqrt{\frac{144\text{ cm}^2}{36\text{ km}^2}} = \frac{12 \times 10\text{ mm}}{6 \times 10^3 \times 10^3\text{ mm}} = \frac{1}{50000}$$

Length of scale = R F x Max length to measure

$$= \frac{1}{50000} \times 10\text{ km} = \frac{1}{50000} \times 10 \times 1000 \times 1000 = 200\text{ mm}$$




R.F. = 1 : 50000

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|---|---|---|---|
| STD | | | |
| COMP | | | PLAIN AND DIAGONAL SCALE |
| ROHITKUMAR GUPTA | | | DRG NO: EGD/MECH/02 |
| 200200119001 | | | SCALE: 1 : 1 |
| 2nd MECH M4/ S21 | | | |

# 3: ENGINEERING CURVES

*PROBLEM NO.1:*

*The foci of an ellipse are 110 mm apart. The minor axis is 70mm long. Determine the length of major axis and draw half ellipse by rectangle method and other half by concentric circle method. Draw tangent and normal to curve at any point.*

*PROBLEM NO.2:*

*Draw ellipse, parabola and hyperbola on the same axis and same directrix. Take distance of focus from the directrix equal to 50 mm and eccentricity for the ellipse, parabola and hyperbola as 2/3, 1 and 3/2 respectively. Plot at least 8 points. Take suitable point on each curve and draw tangent and normal to the curve at that point.*

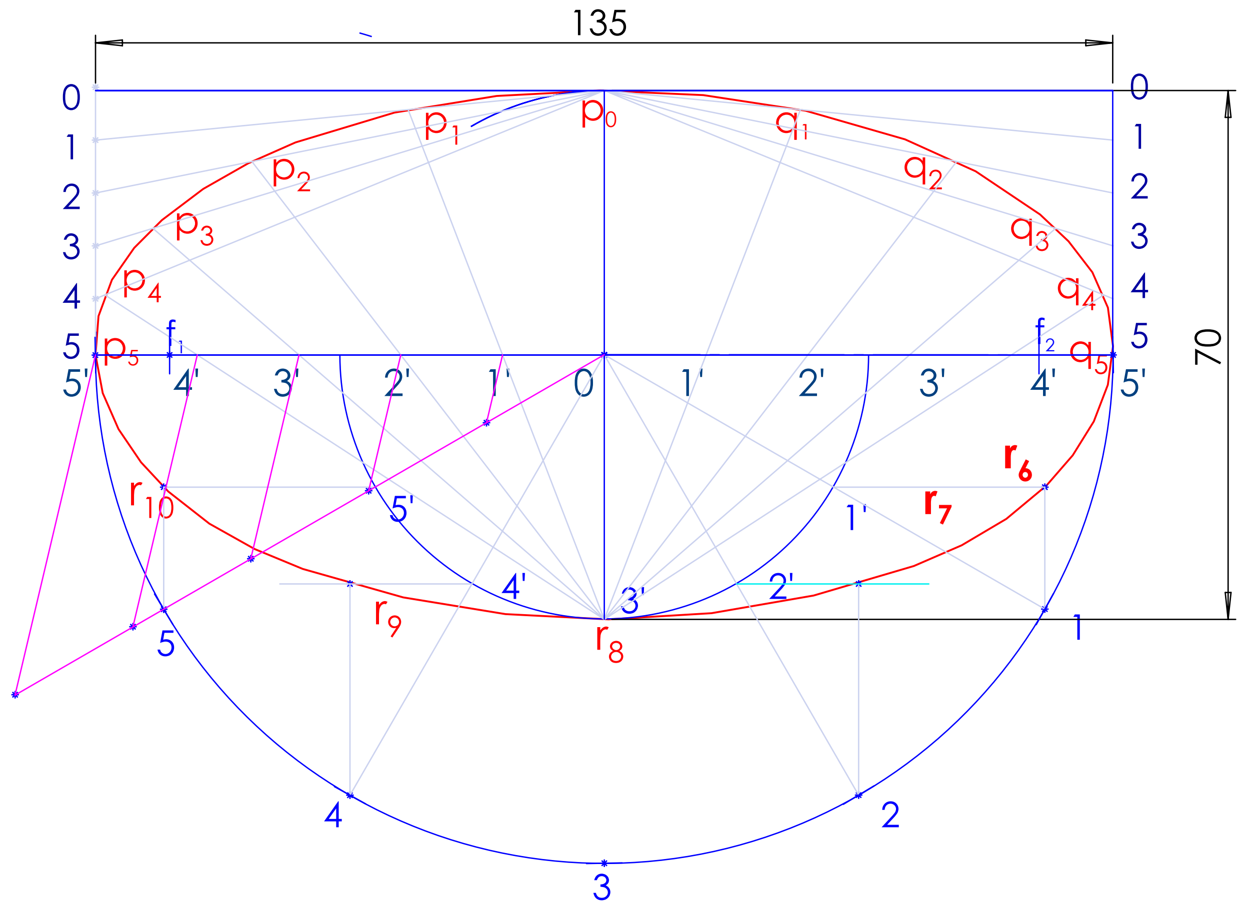
*PROBLEM NO.3:*

*Construct the involute of circle of 30mm diameter for one turn. Draw tangent and normal to the involute at any point of it.*

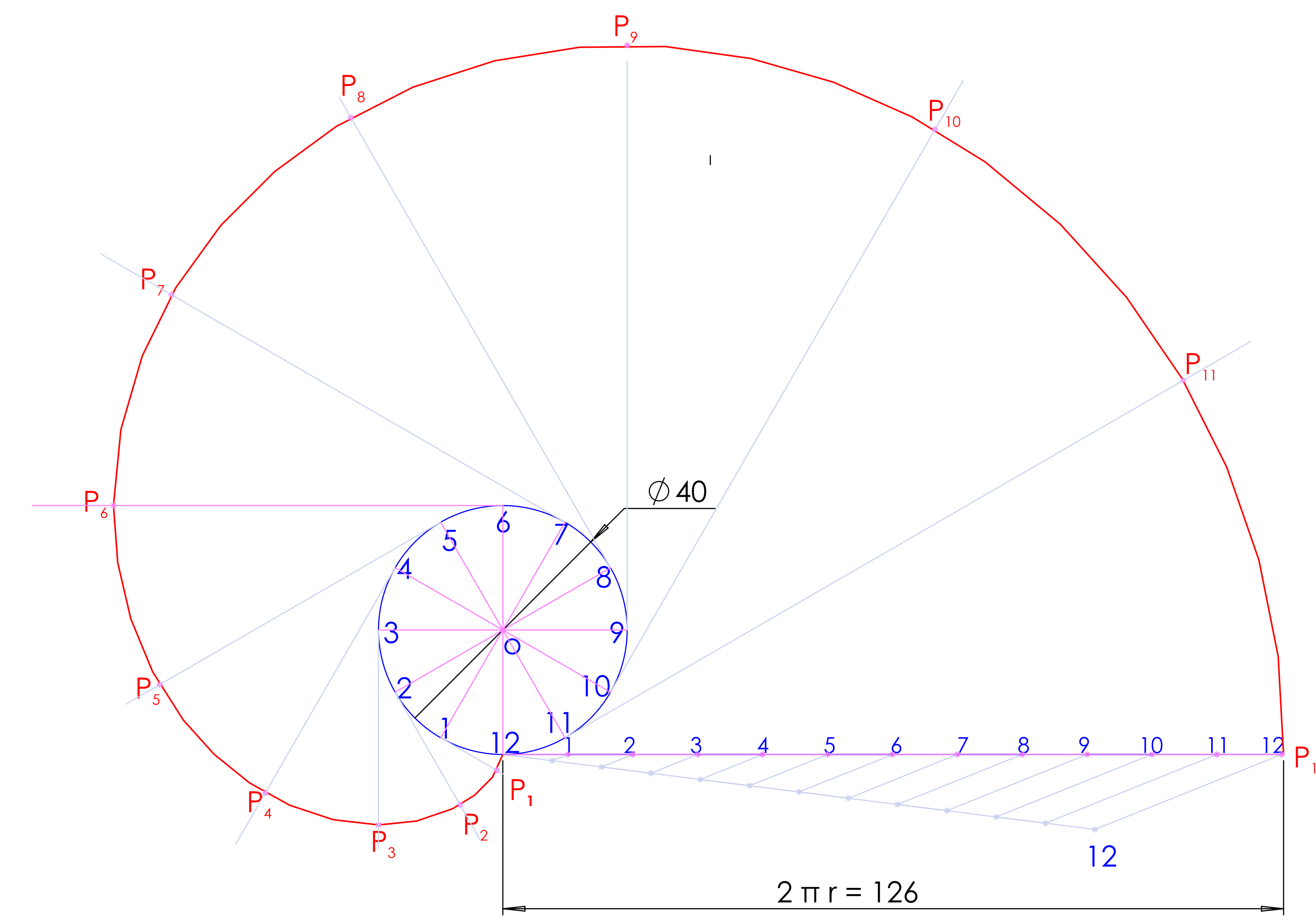
*PROBLEM NO.4:*

*Draw a cycloid for a rolling circle of 60 mm diameter rolling along a straight line without slipping. Take initial position of the tracing point at the bottom of the vertical Centre line of the rolling circle. Draw tangent and normal to the curve at a point 35mm above the directing line.*

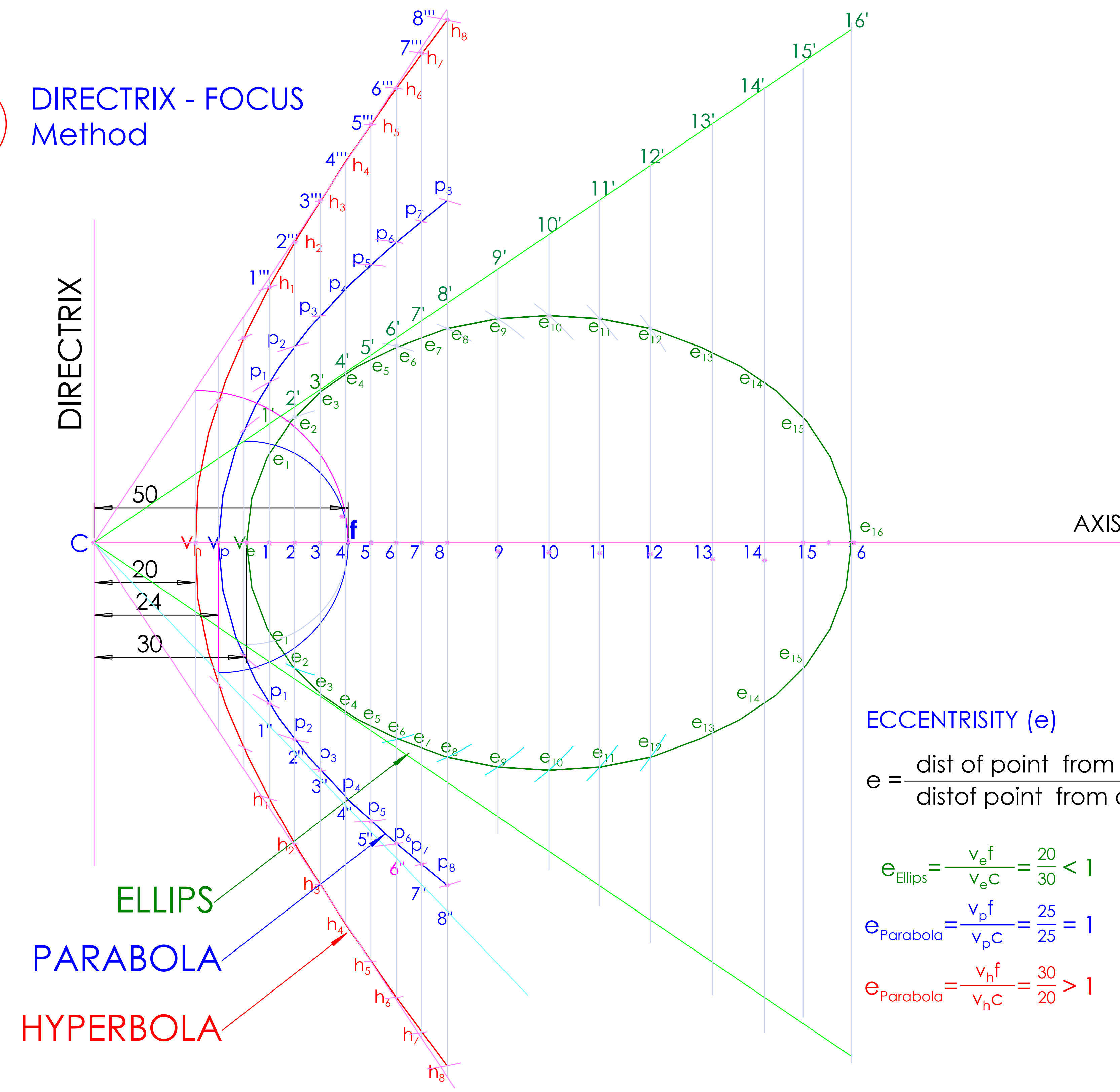
## 1 ELLIPS Ractangle & Concentric Circle method




## 3 INVOLUTE OF CIRCLE




## 2 DIRECTRIX - FOCUS Method




ECCENTRICITY (e)

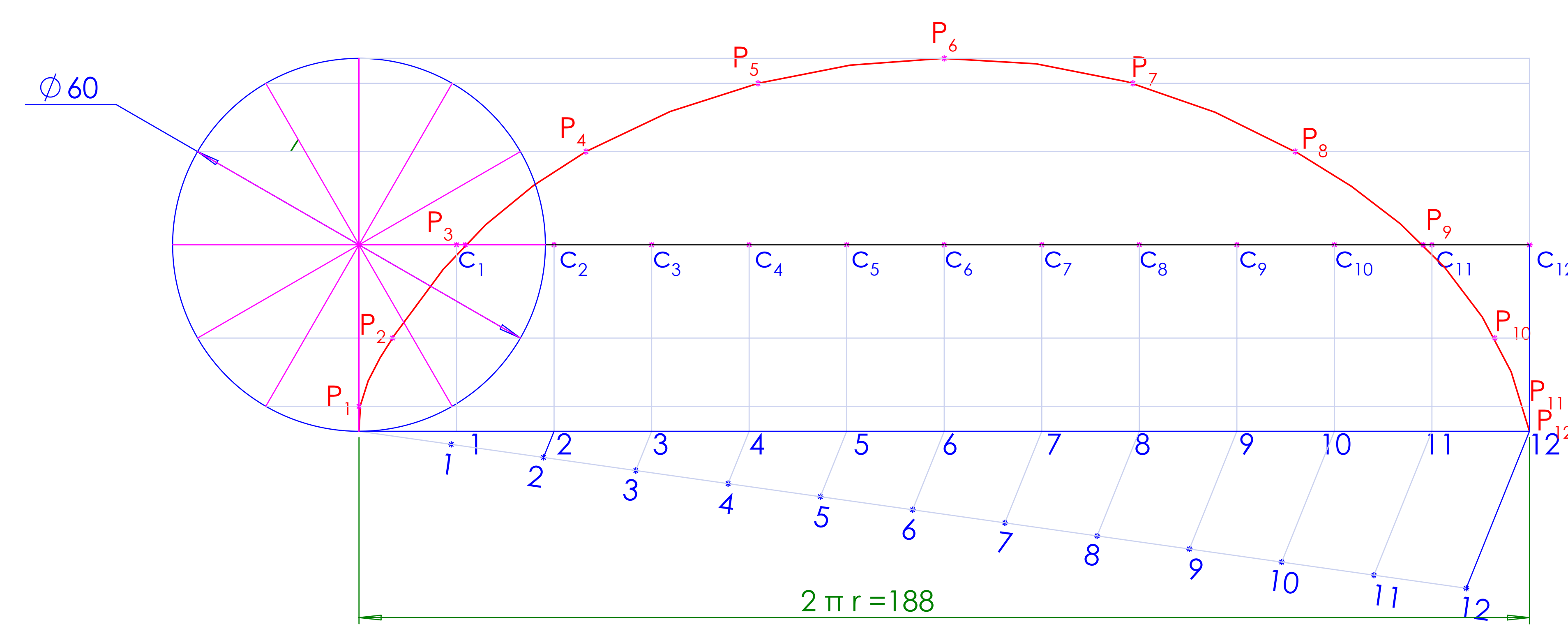
$$e = \frac{\text{dist of point from focus}}{\text{distof point from directrix}}$$

$$e_{Ellips} = \frac{v_e f}{v_e C} = \frac{20}{30} < 1$$

$$e_{Parabola} = \frac{v_p f}{v_p C} = \frac{25}{25} = 1$$

$$e_{Parabola} = \frac{v_h f}{v_h C} = \frac{30}{20} > 1$$

## 4 CYCLOID

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| | SIGN | DATE | GOVERNMENT ENGINEERING COLLEGE, | |
|---|---|---|---|---|
| STD | | | RAJKOT | |
| COMP | | | ENGINEERING CURVES | |
| SADIYA SANABHAI | | | DRG NO: | EGD/MECH/04 |
| 210200119067 | | | SCALE: | 1 : 1 |
| 2nd MECH M4/ S22 | | | | |

# 4: PROJECTION OF POINTS AND LINES

*PROBLEM NO.1:*

*Draw the projections for the below points. Take a single reference line. The distance between the projectors is 20mm.*

*a) Point 'A' is 20mm above HP and 30mm in front of P.*
*b) Point 'B' is 20mm below Hp and 40mm behind VP.*
*c) Point 'P' is 10mm above HP and 30mm Behind VP.*
*d) Point 'C' is 45mm below HP and 35mm in front of VP.*
*e) Draw the projections of a point 'Q' lies on both HP and VP.*
*f) The point 'K' lies on HP and 10mm in front of VP.*

*PROBLEM NO.2:*

*The distance between the end projectors of a straight-line PQ is 130mm. The end P is 40mm below HP and 25mm in front of the VP. Q is 75 mm above HP and 30 mm behind VP. Draw its projections. Find TL of the line.*

*PROBLEM NO.3:*

*A line AB has its end point A 15 mm above H.P. and 25 mm Infront of V.P. Line AB is making an angle of 20° to the H.P. Length of the line in the T.V. is 90mm. End B of the line is equidistant from both the principal planes. Find the true length of the line and the angle made by the line with V.P.*

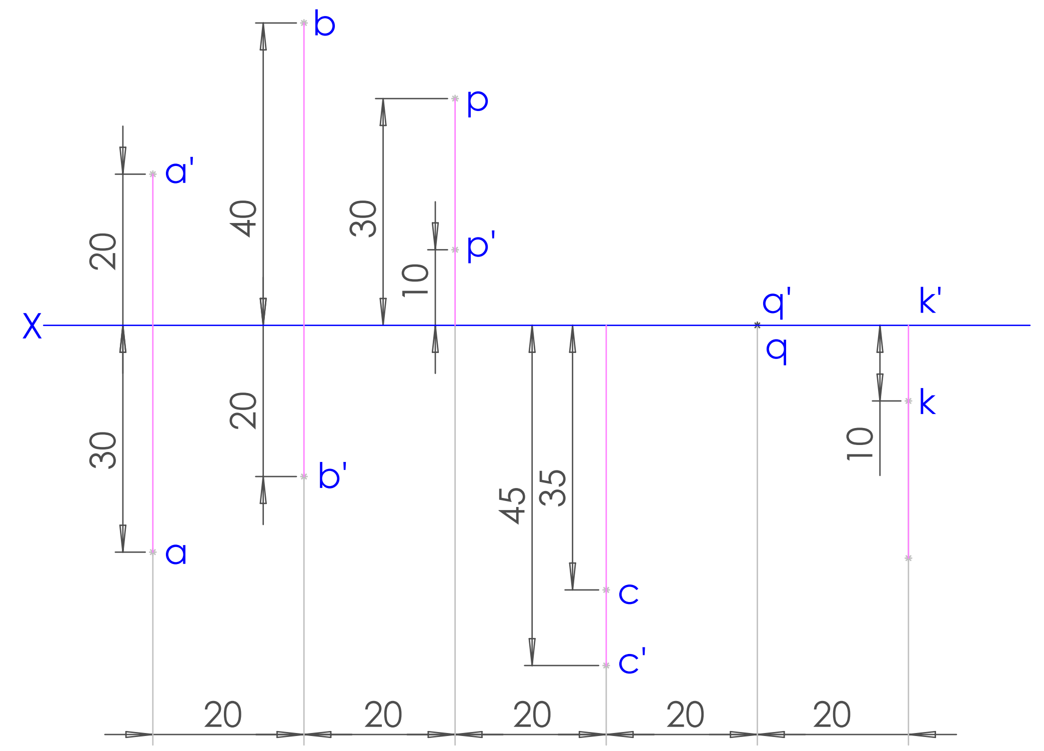
*PROBLEM NO.4:*

*A straight-line AB 80 mm long is inclined at 30° to the HP and at 45° to the VP. Its mid-point C is in the VP and 18 mm above the HP, while its end A is in the third quadrant, and the end B is in the first quadrant. Draw its projections.*

*PROBLEM NO.5:*

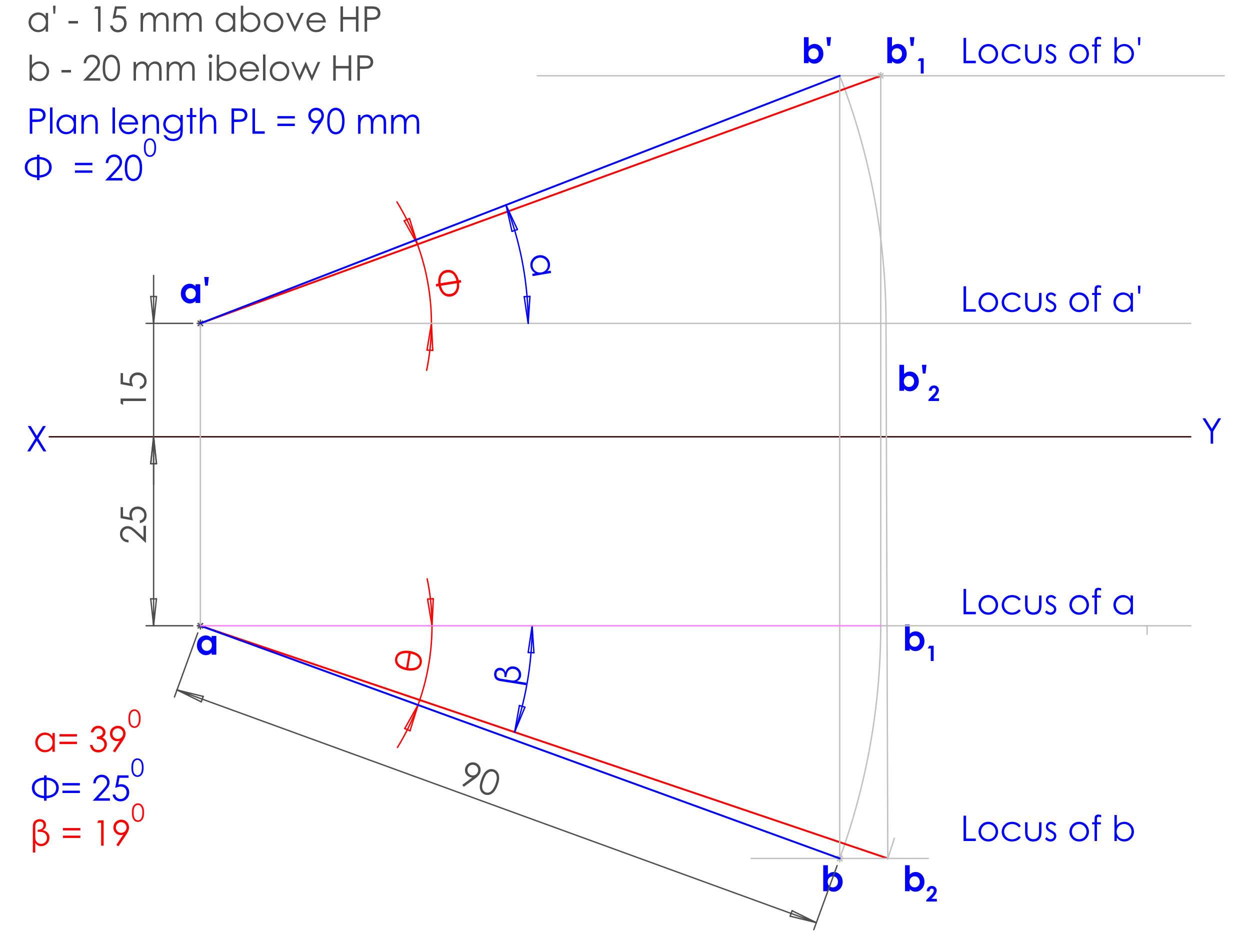
*Plan and elevation of a line AB, 80 mm long, measure 60 mm and 72 mm respectively. End A of the line is in HP and end B is in VP. Draw its projections, assuming the line to lie in first quadrant.*

1




3

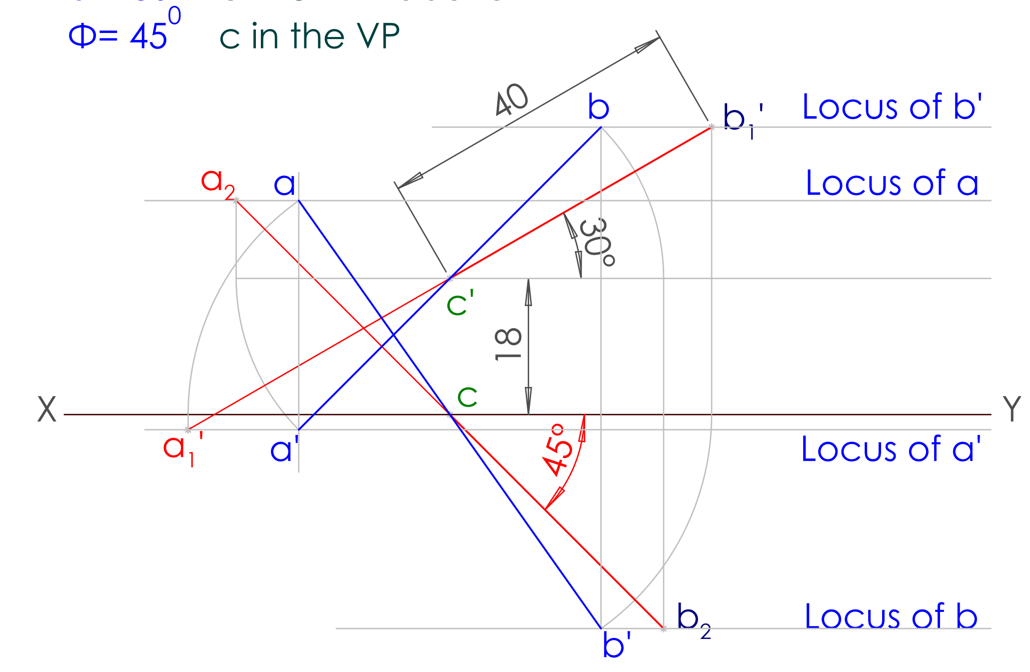
a' - 15 mm above HP
b - 20 mm ibelow HP
Plan length PL = 90 mm
$\Phi = 20^0$




$\alpha = 39^0$
$\Phi = 25^0$
$\beta = 19^0$

4

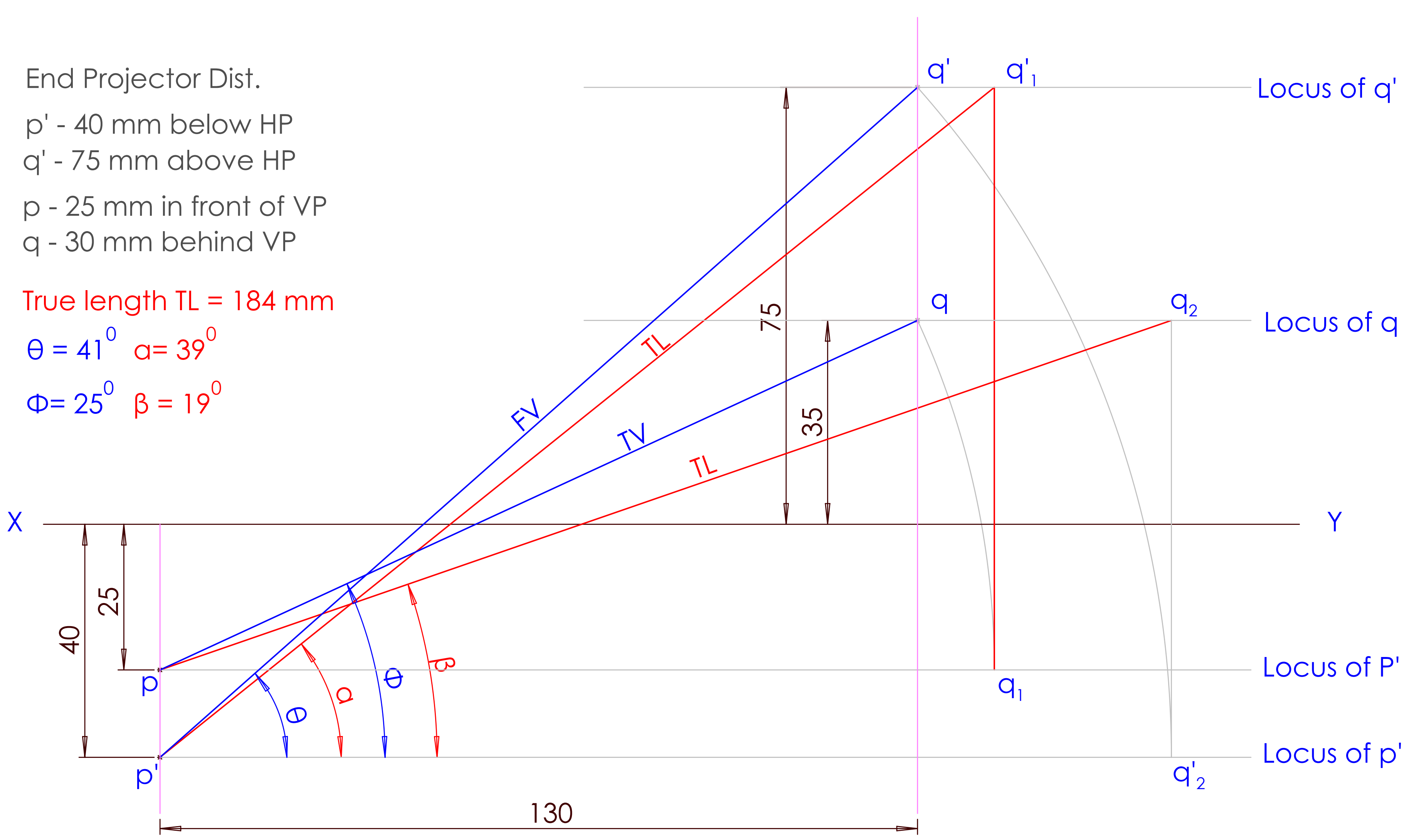
True length TL = 80 mm
$\theta = 30^0$ c'=18 mm above HP
$\Phi = 45^0$ c in the VP




2

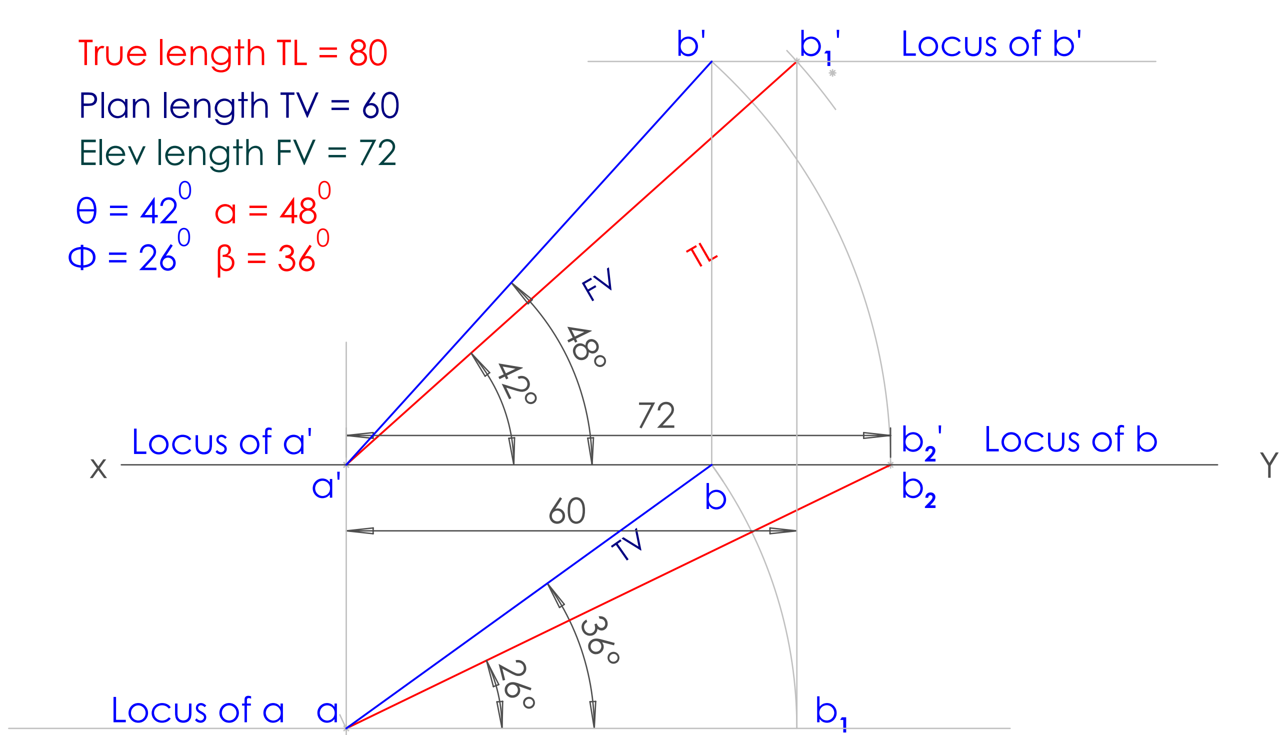
End Projector Dist.
p' - 40 mm below HP
q' - 75 mm above HP
p - 25 mm in front of VP
q - 30 mm behind VP

True length TL = 184 mm
$\theta = 41^0$ $\alpha = 39^0$
$\Phi = 25^0$ $\beta = 19^0$




5

True length TL = 80
Plan length TV = 60
Elev length FV = 72
$\theta = 42^0$ $\alpha = 48^0$
$\Phi = 26^0$ $\beta = 36^0$

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|---|---|---|---|
| STD | | | |
| COMP | | | PROJECTIONS OF POINTS AND LINE |
| CHUDASAMA YASHPALSINH | | | |
| 220200119012 | | | DRG NO: EGD/MECH/05 |
| 2nd MECH M4/ S22 | | | SCALE: 1 : 1 |

# 5: PROJECTION OF PLANES

*PROBLEM NO.1:*

*A regular pentagon of 30mm side has one side parallel to the V.P. and making an angle of 40° with the H.P. the plane surface of the pentagon makes 35° the V.P. Draw its projections.*

*PROBLEM NO.2:*

*A rhombus of negligible thickness is having its diagonals 100 mm and 50 mm long. Draw the projections of the rhombus when the longer diagonal is inclined at 30° to the Horizontal Plane and 30° to Vertical Plane.*

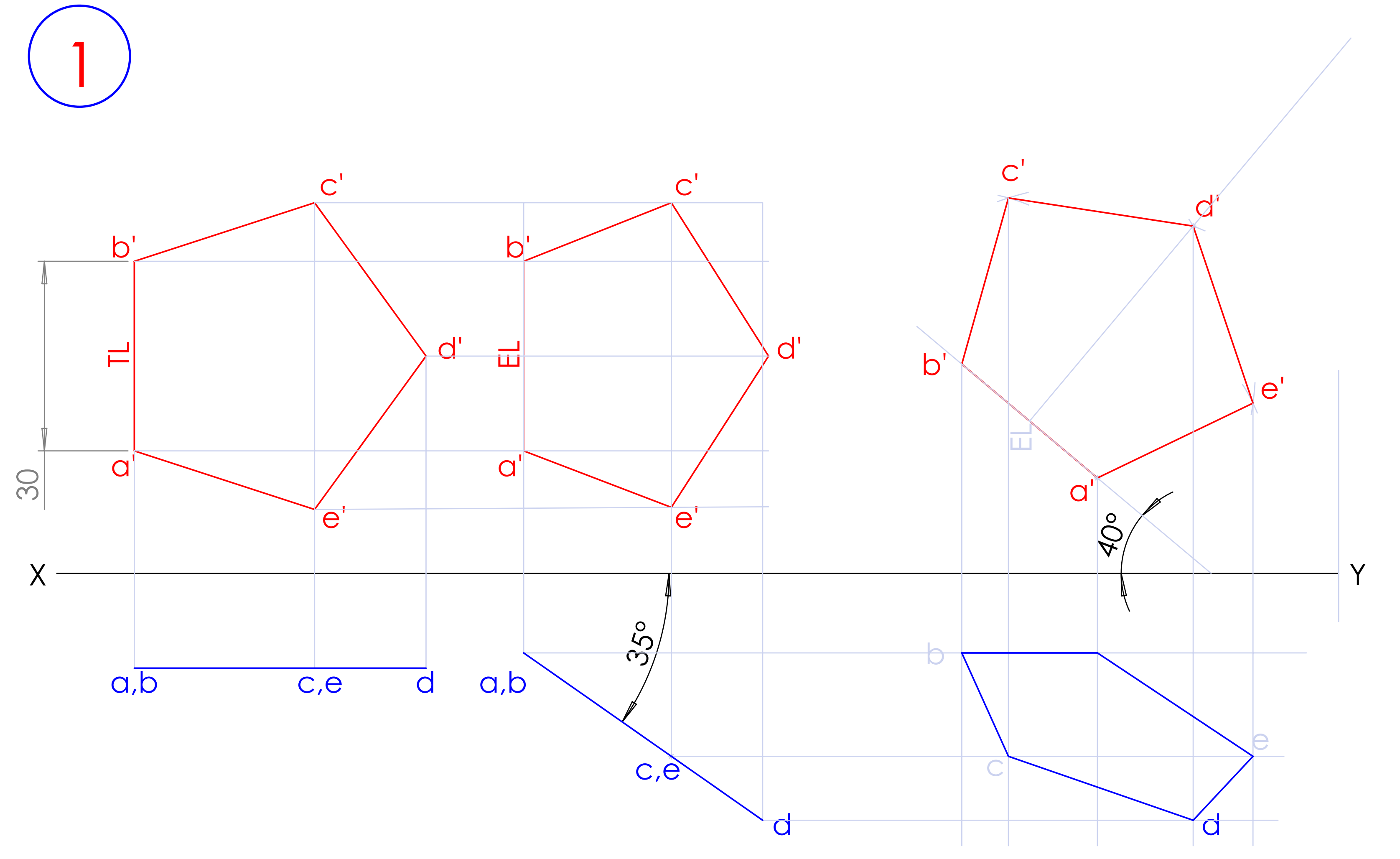
*PROBLEM NO.3:*

*A circular plane having the diameter 50 mm is resting with point A of its periphery on H.P. The surface of the plane is inclined to H.P. such that the plan of the plane becomes an ellipse with minor axis 30mm. Draw the projection of the plane when the plan of the diameter through point A is inclined at 30° to V.P.and the center of the plane is 40mm from V.P.Find the inclination of the plane with H.P.*

*PROBLEM NO.4:*

*An isosceles triangular plate ABC has its base 45 mm and altitude 60 mm. It is so placed that the front view is seen as an equilateral triangle of 45 mm side and (i) base is inclined at 45° to HP, (ii) side is inclined at 45° to HP. Draw its plan when its corner A is on HP.*

1

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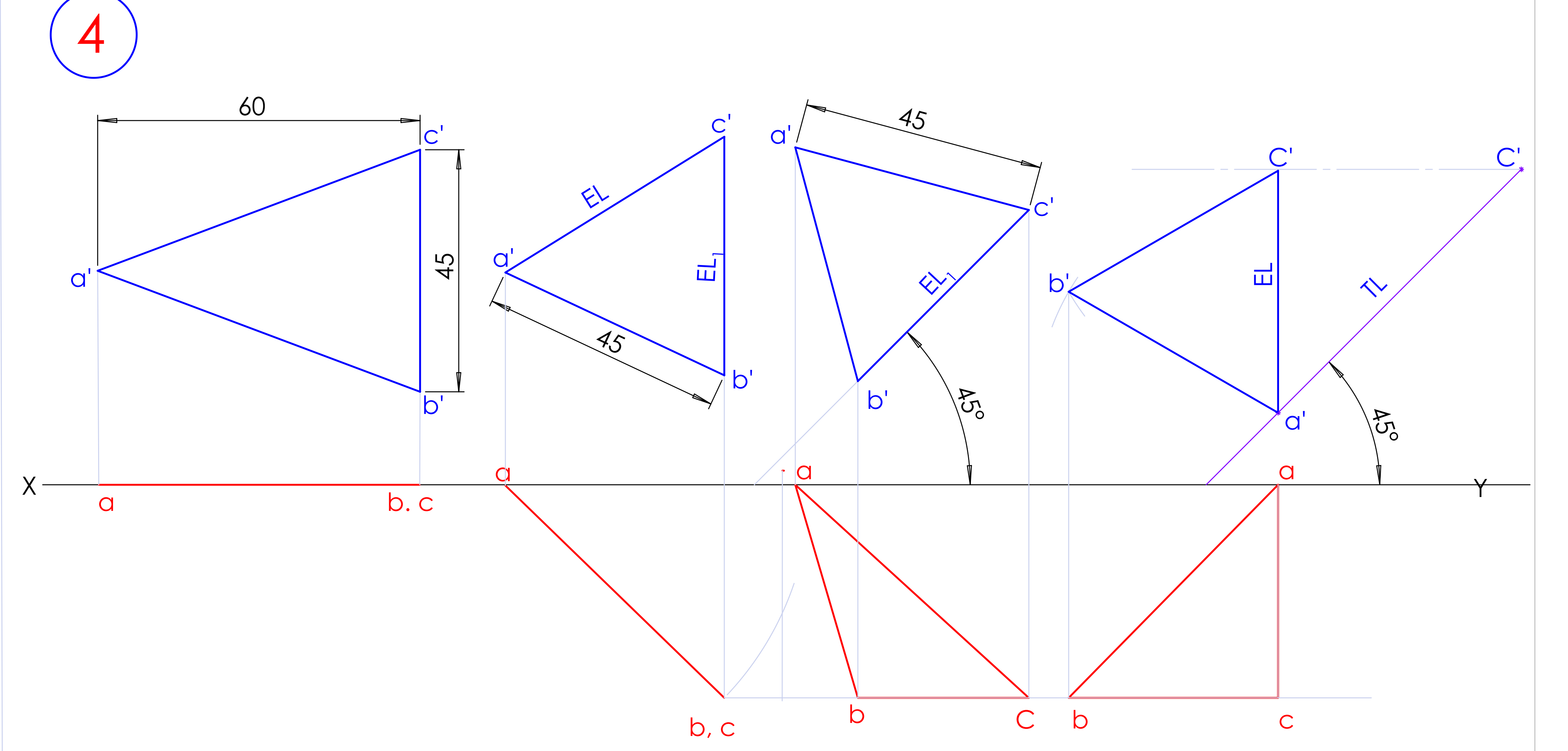


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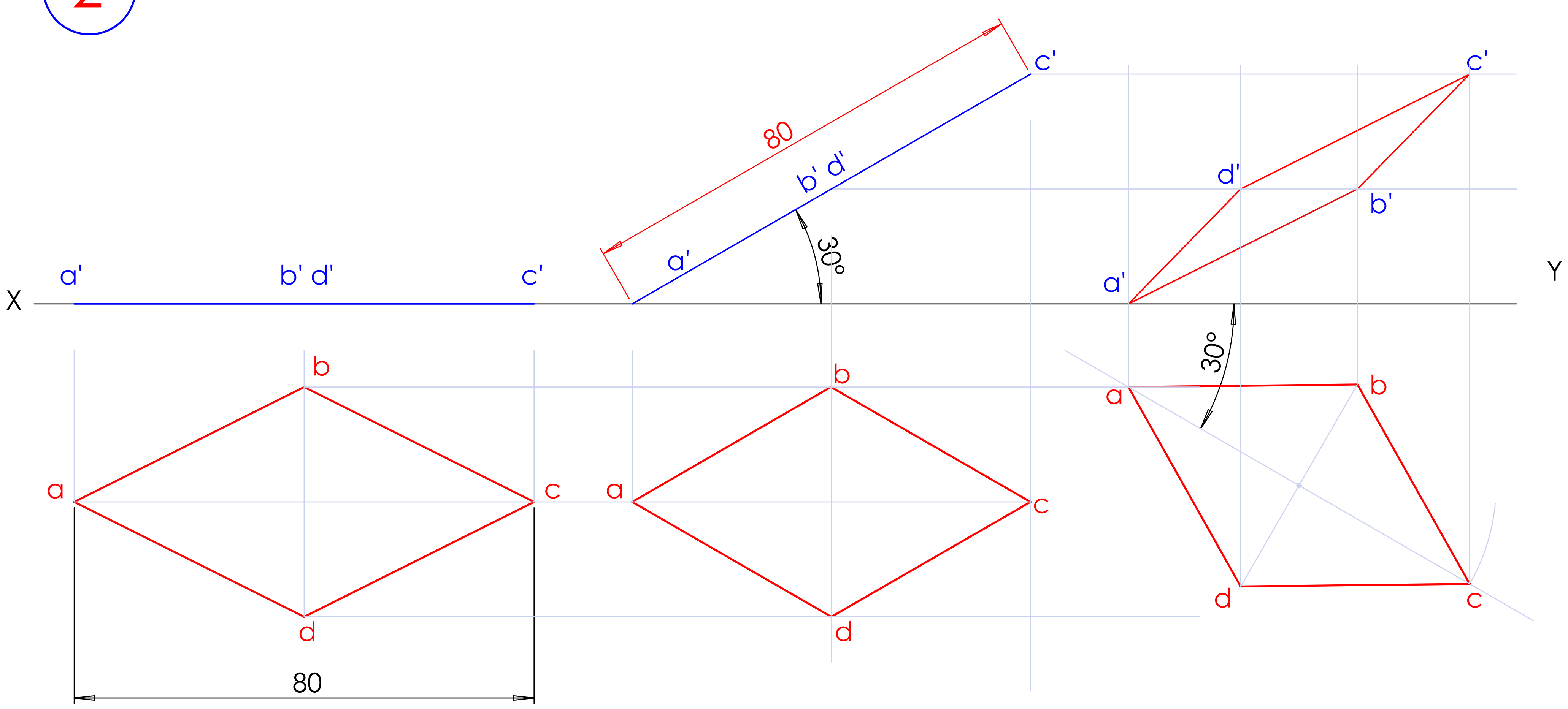


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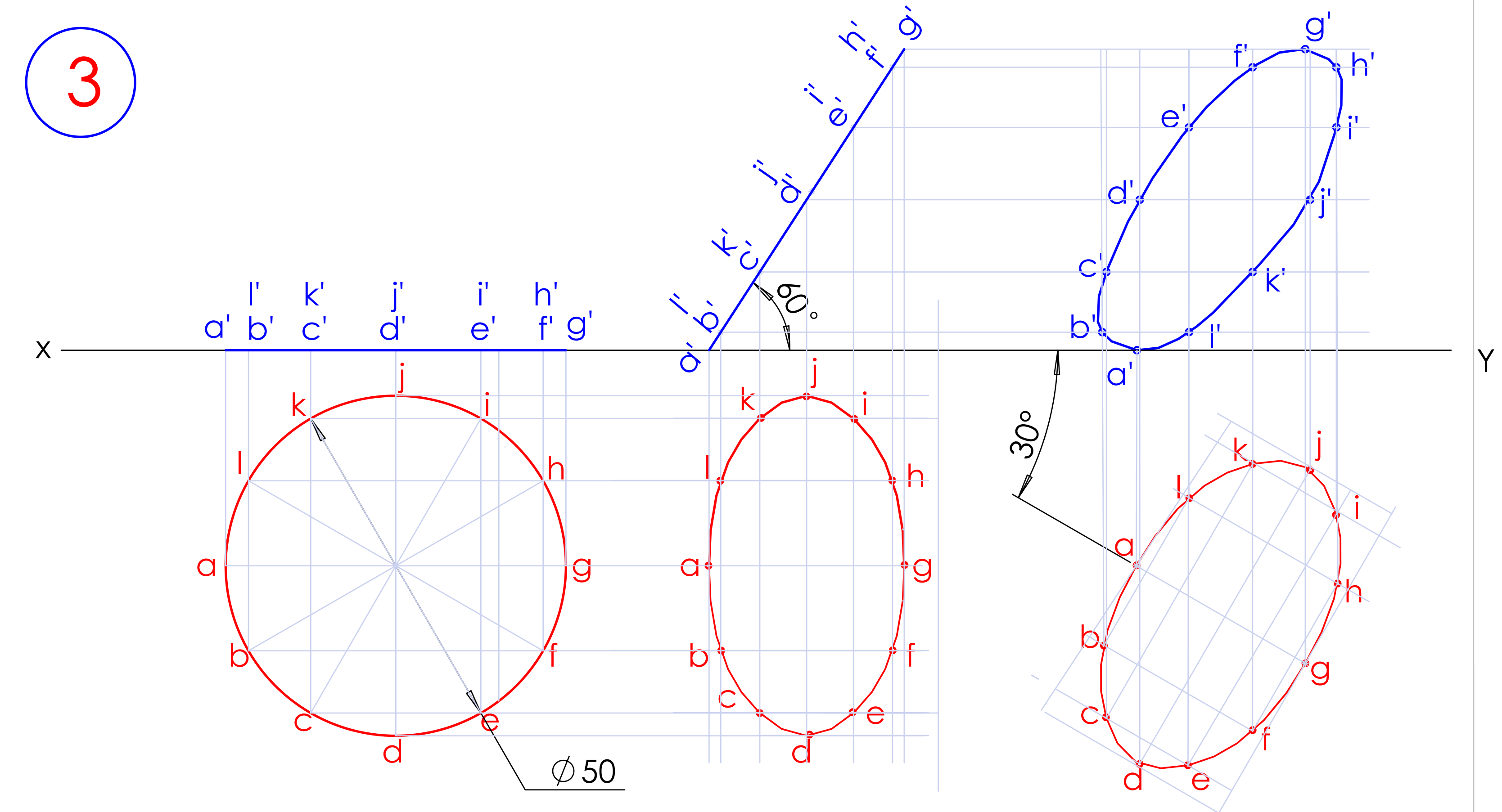


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|---|---|---|---|---|
| STD | | | PROJECTIONS OF PLANE | |
| COMP | | | | DRG NO: EGD/MECH/06 |
| MORI HARDIK | | | | SCALE: 1 : 1 |
| 210200119005 | | | | |
| 2nd MECH M4/ S22 | | | | |

# 6: PROJECTION OF SOLID, SECTION OF SOLID

*PROBLEM NO.1:*

*A hexagonal prism side of base 30 mm & axis length 60 mm is resting on one of its base edges on HP such that its axis inclined at 45° to HP and the side on which it is resting is inclined at 30° to VP. Draw the projections.*

*PROBLEM NO.2:*

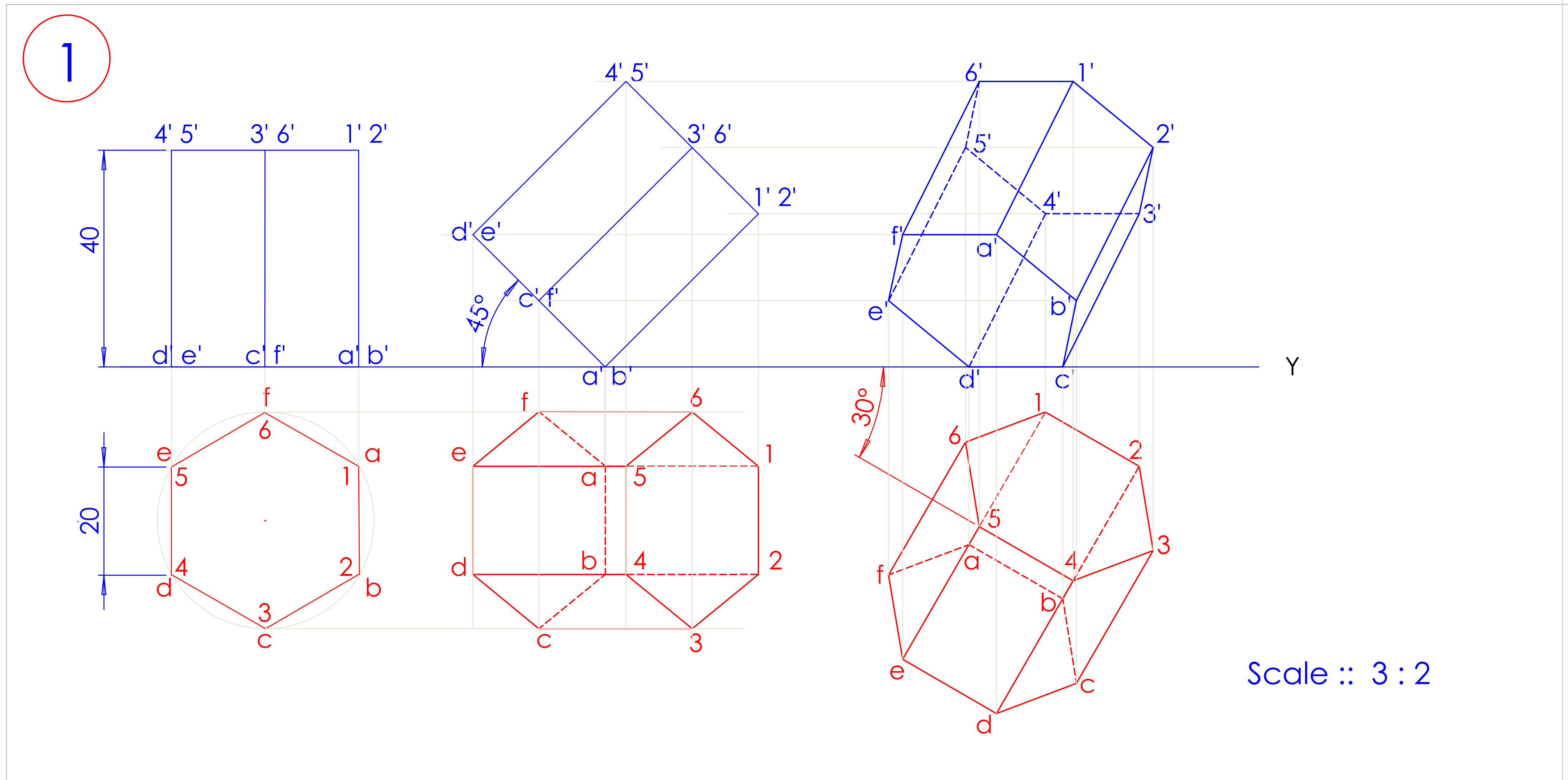
*A tetrahedron of 50 mm long edges is resting on one edge on HP while one triangular face containing this edge is vertical and 45° inclined to VP. Draw projections.*

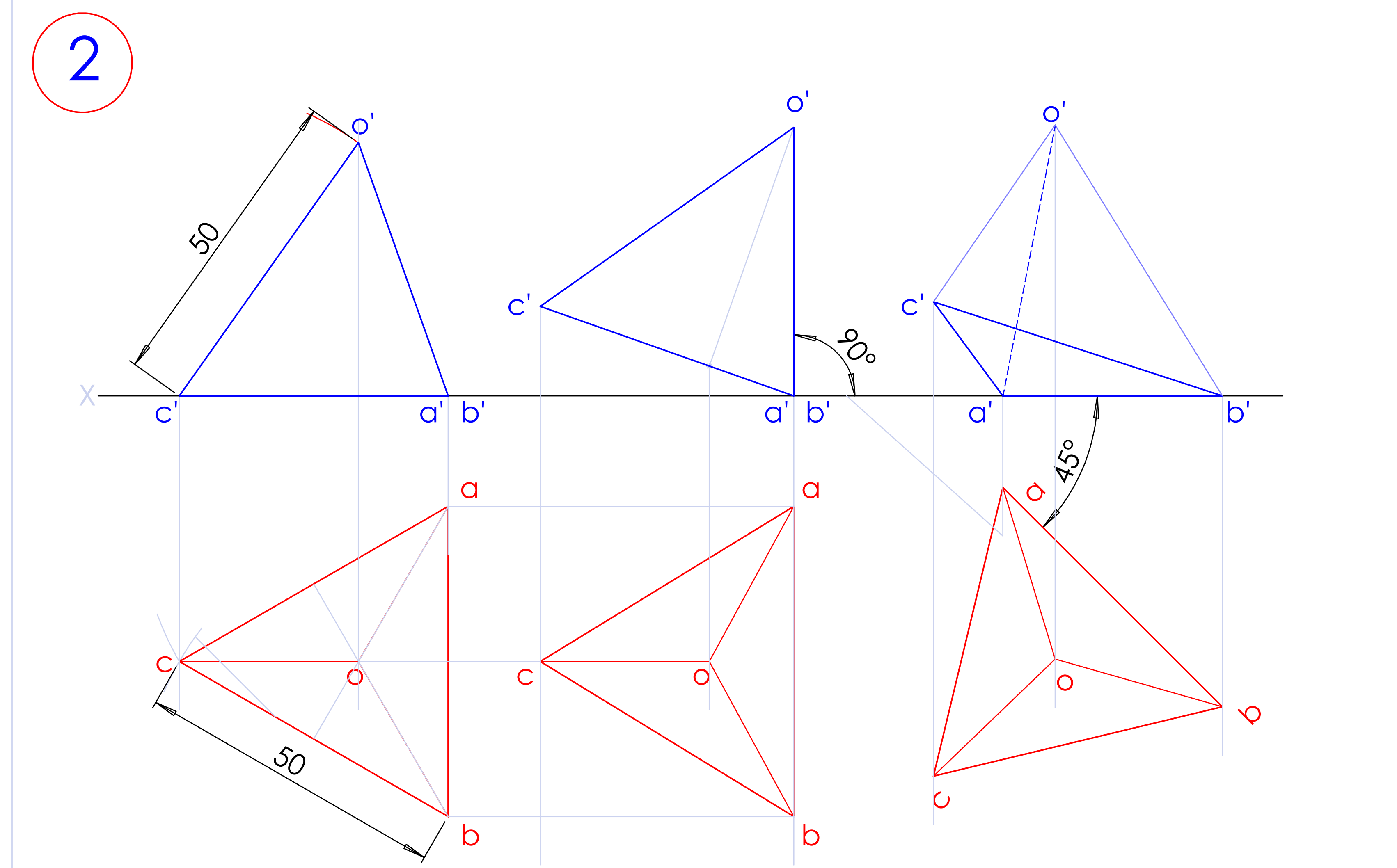
*PROBLEM NO.3:*

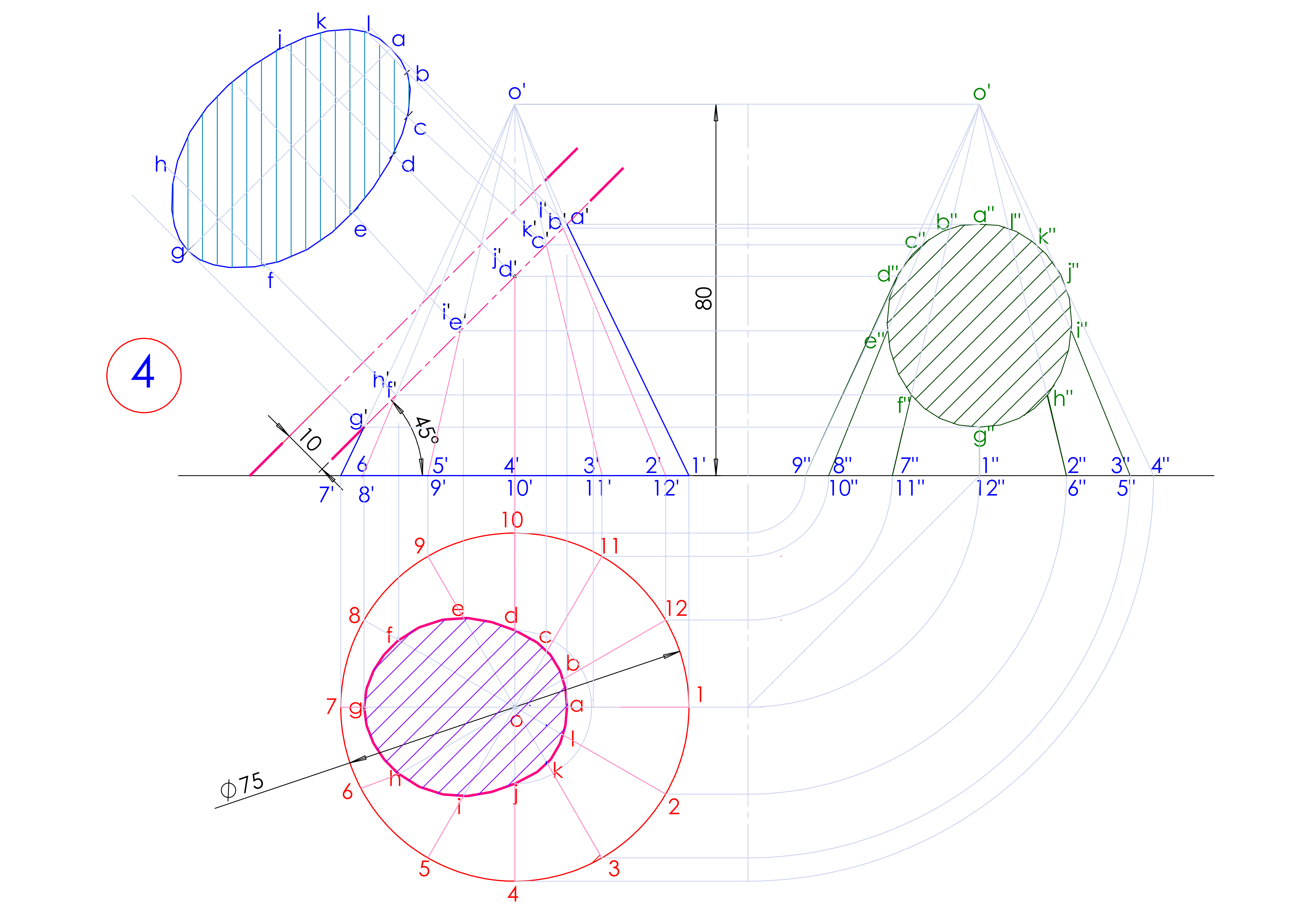
*A hexagonal prism, edge of base 20 mm and axis 50 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the right corner of the top face of the prism. (i) Draw the sectional top view. (ii) Draw the sectional side view*

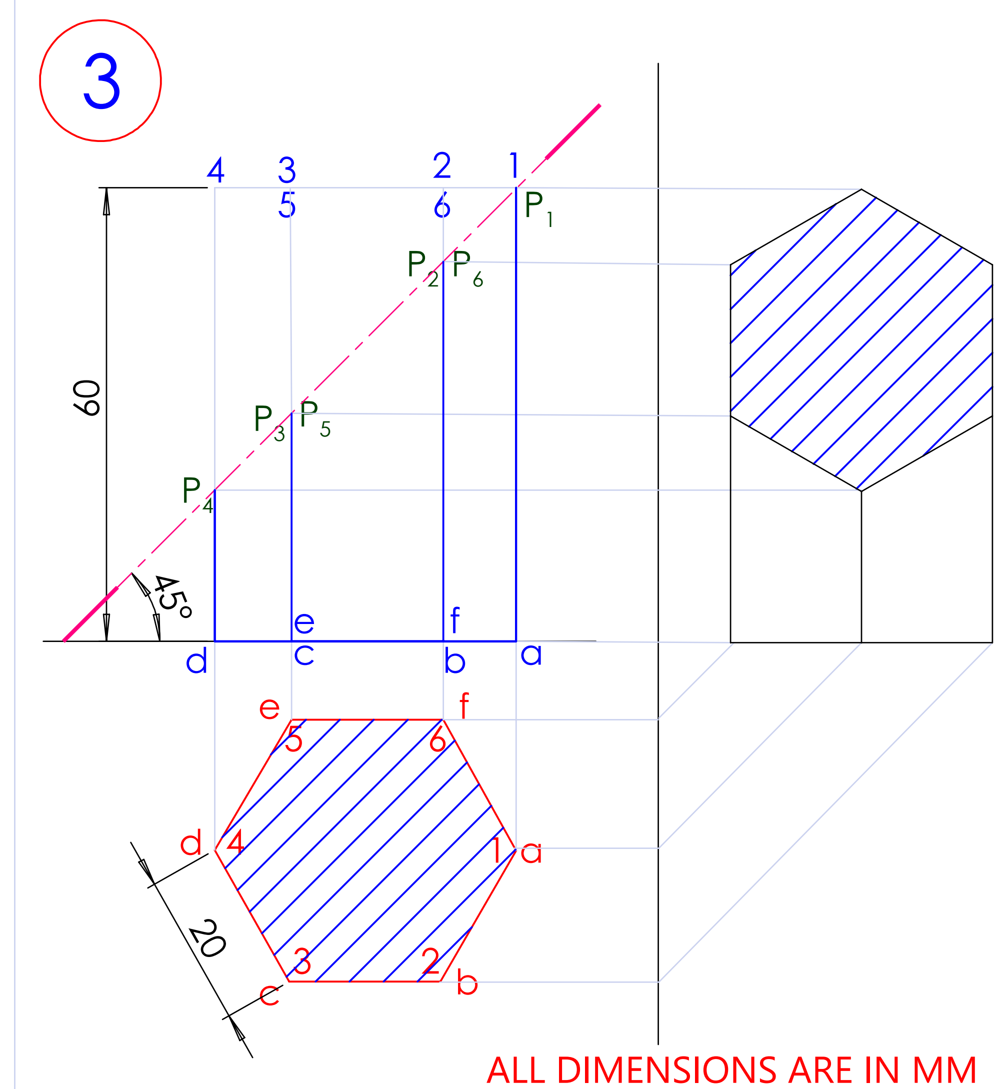
*PROBLEM NO.4:*

*A Cone base 75 mm diameter and axis 80 mm long is resting on its base on H.P. It is cut by a section plane perpendicular to the V.P., inclined at 45º to the H.P. and cutting the axis at a point 35 mm from the apex. Draw the front view, sectional top view, sectional side view and true shape of the section*

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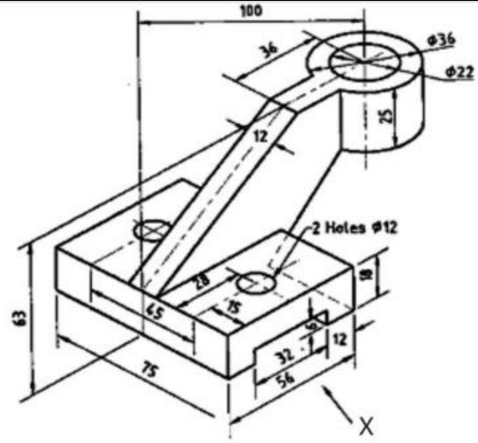
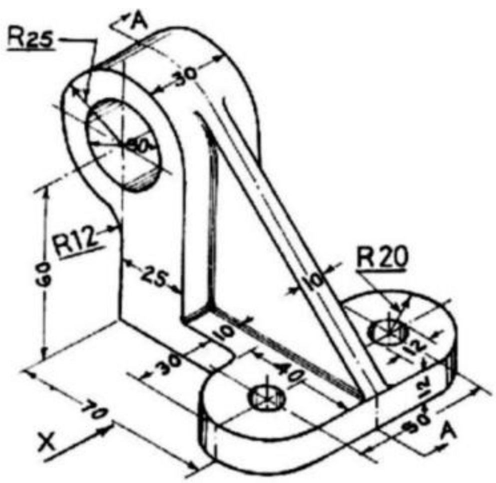
|---|---|---|---|
| STD | | | |
| COMP | | | PROJECTIONS & SECTION OF SOLIDS AND DEVLOPMENT OF SURFACES |
| HARDIK VEKARIYA | | | |
| 210200119025 | | | DRG NO: EGD/MECH/07 |
| 2nd MECH M4/ S22 | | | SCALE: 1 : 1 |

# 7: ORTHOGRAPHIC PROJECTIONS

*PROBLEM NO.1:*

*Draw the front view looking from X-direction plan and LH side view of a given object using first angle projection method.*




*PROBLEM NO.2*

*Figures shows pictorial view of an object Draw the following view:*

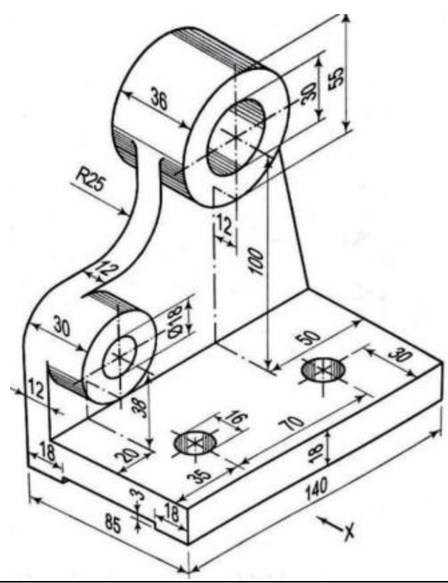
*(a) Sectional elevation from x*

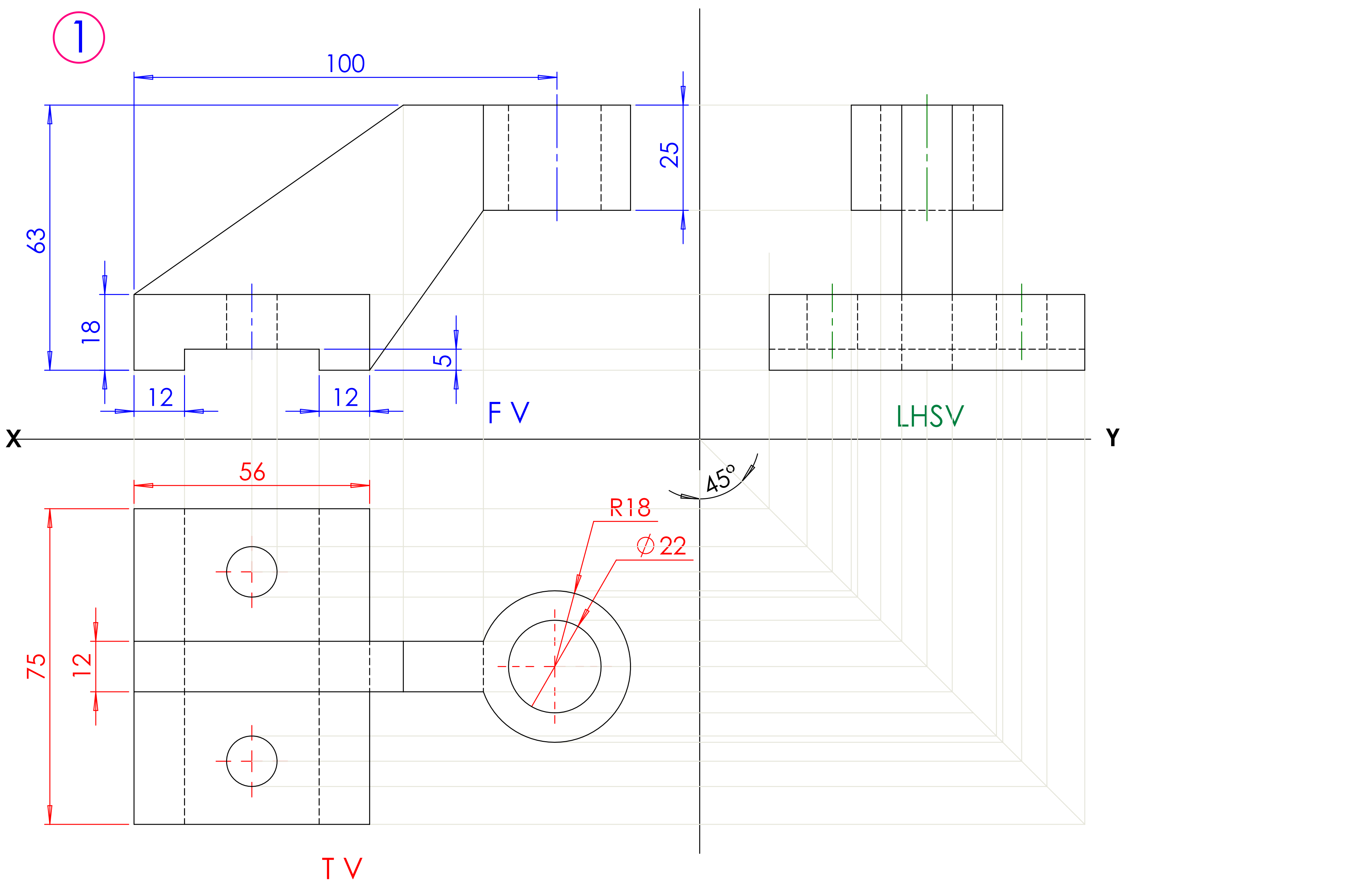
*(b) plan and*

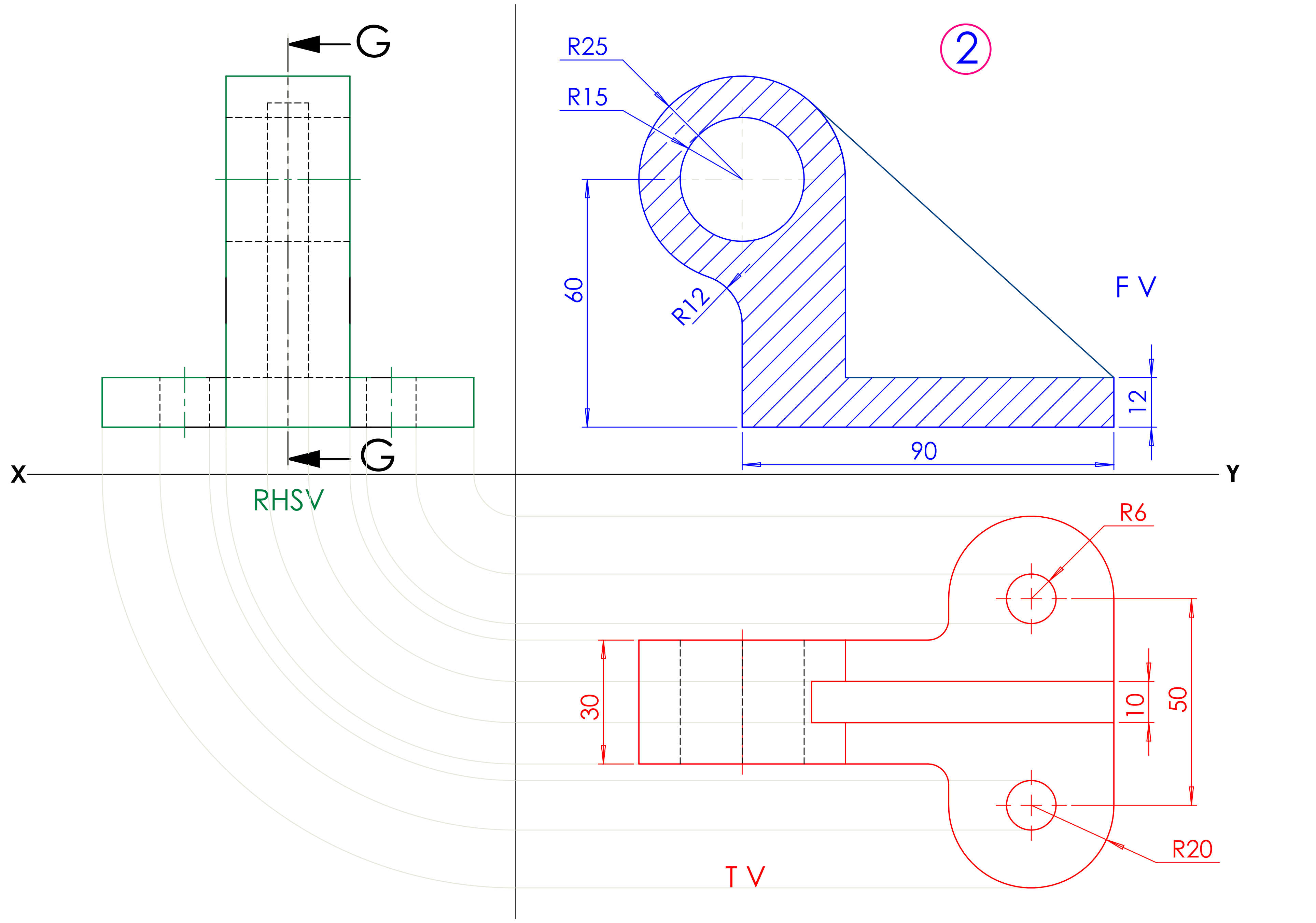
*(c) right hand side view*

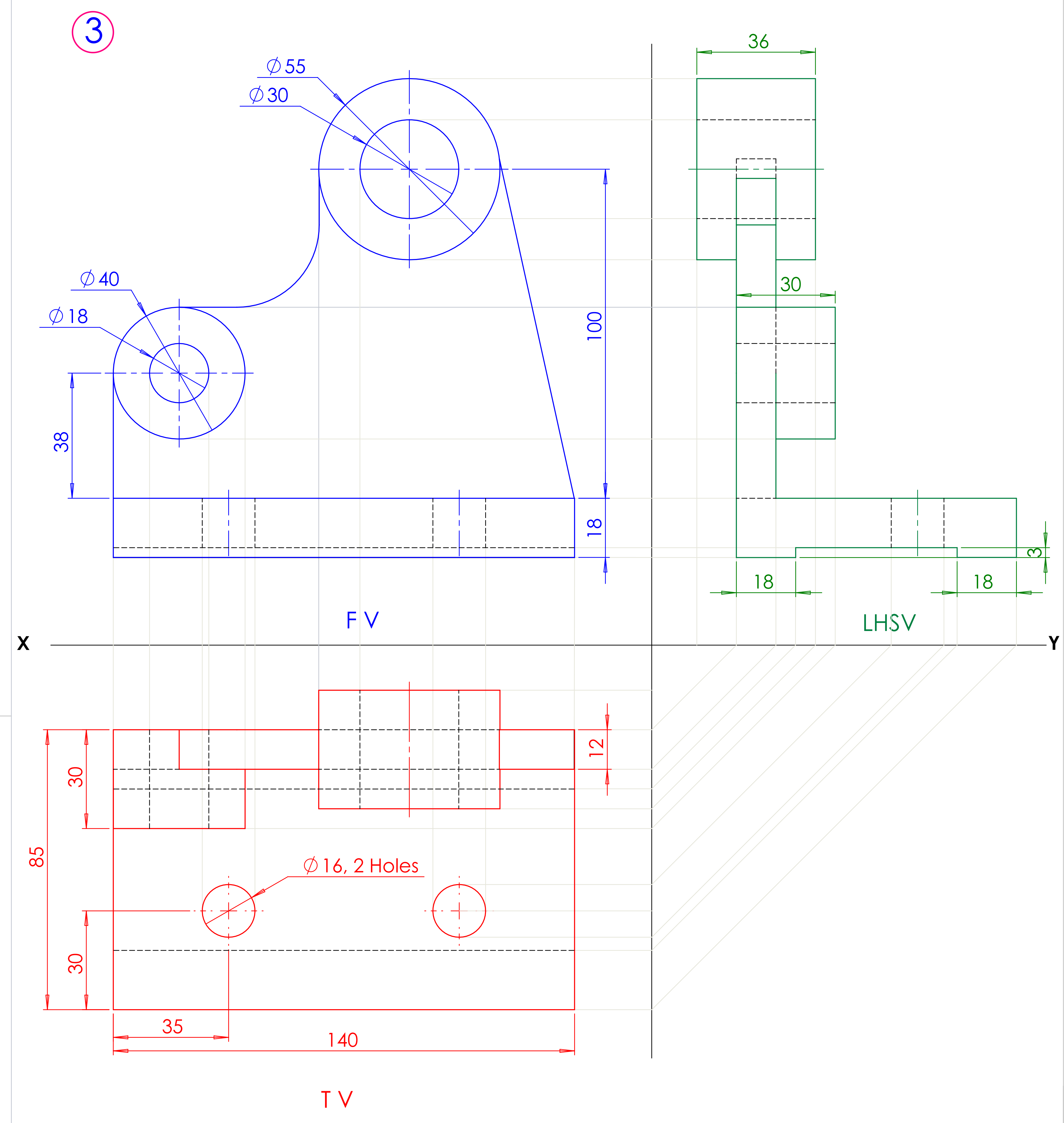
*PROBLEM NO.3*

*Draw the, front view looking from X - direction, plan and LH side view of a given object using first angle projection method.*



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|---|---|---|---|---|
| STD | | | ORTHOGRAPHICS PROJECTIONS | |
| COMP | | | | |
| DHYEY VALA | | | DRG NO: | EGD/MECH/08 |
| 210200119029 | | | SCALE: | 4 : 5 |
| 2nd MECH M4/ S22 | | | | |